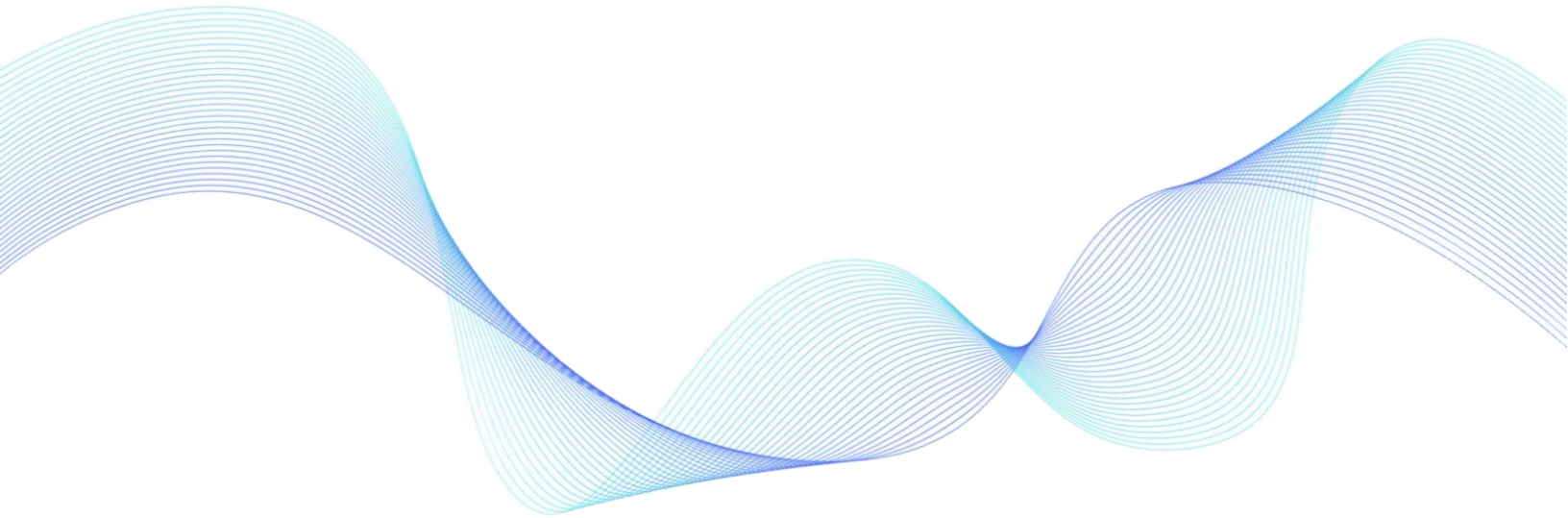
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Leveraging Automated Machine Learning for Scalable Predictive Analytics and Decision Support Systems

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Keywords: Machine Learning, Predictive Analytics, Decision Support Systems, Automated Machine Learning (AutoML), Data-Driven Decision Making, Web-Based AutoML



100675768@unimail.derby.ac.uk

Abstract

Machine Learning has become one of the most significant technological advancements in data analytics due to its ability to extract information from rapidly growing datasets. Large volumes of data have been generating since the expansion of digital technologies such as online platforms and cloud computing. This generation of data covers multiple fields like business, healthcare, finance and other industries. As a result, businesses rely heavily on predictive analytics and intelligent systems for support strategies and operational decision-making processes. Decision Support Systems combined with Machine Learning enable businesses to identify patterns, predict outcomes and increase the rate at which decisions are made in order to surge marketing success.

In modern times, Artificial Intelligence has a significant role in Decision Support Systems since it introduces automated and scalable environments capable of handling complex data with minimal intervention required. This integration has re-evaluated the term Automated Decision Support. Nevertheless, challenges regarding correct interpretability, data handling and balance between automation and human intervention are continuously influencing the use and effectiveness of such systems.

The research conducted in this study aims to contribute to the transparency of Human-AI collaboration as well as the integration of Machine Learning for decision support and business value. In addition, this dissertation presents the design and implementation of an interactive web-based Automated Machine Learning (AutoML) system for predictive analytics and decision support. This implementation follows a specific workflow and step-by-step methodology that has been supported by research of the industry. Testing on the system is conducted using three different datasets from which both regression and classification analyses take place. Although the application presented is not finalized, further suggestions and reflections are discussed that can support additional research on topics covered in this study.

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Table of Contents

Abstract 2
Acknowledgments 2
1 Introduction 5
1.1 Machine Learning Background and Research Motivation 5
1.2 Problem Statement and Research Gaps 6
1.3 Aims and Objectives 7
2 Literature Review 7
2.1 Foundations of Machine Learning 7
2.1.1 Supervised and Unsupervised Learning 8
2.1.2 Supervised Learning Application in Predictive Tasks 9
2.2 Predictive Analytics and Pattern Extraction 10
2.2.1 The Predictive Analytics Process 10
2.2.2 Pattern Extraction and Sequential Pattern Mining 11
2.3 Decision Support Systems 12
2.3.1 The Evolution of Decision Support Systems 12
2.3.2 Design Methodologies and Implementation 12
2.3.3 Integration of Machine Learning in Decision Support 14
2.4 Automation in Machine Learning and Decision Support 15
2.4.1 Automated Machine Learning (AutoML) 15
2.4.2 Explainable AI in Machine Learning and Decision Support Systems 19
2.5 Considerations of Human–AI Integration in Decision Support Systems 22
2.6 Review Summary 24
3 DataPilot: Design, Methodology and Evaluation of the Proposed System 24
3.1 Reference Architecture 25
3.2 Proposed System Design 26
3.3 System Implementation Using Reflex 29
3.3.1 Code Parts of Implementation 30
3.4 User Interface 34
3.5 System Testing and Evaluation 38
3.5.1 Regression test 38
3.5.2 Classification Test 43
3.5.3 Train/Test Split Testing 48

3.5.4 Data Validity Test 55
3.5.5 Insights of Testing Results 58
4 Conclusion 58
4.1 Objectives Achieved 59
4.2 Future Work 59
5 References 61

1 Introduction

1.1 Machine Learning Background and Research Motivation

Machine Learning (ML) emerged from the fields of computer science, statistics and pattern recognition as researchers sought out methods that would allow computers to learn from data instead of relying on programmed instructions (Bishop, 2006). Early developments of ML were connected to artificial intelligence research during the mid to late twentieth century where algorithms designed to imitate aspects of human learning were implemented (Carbonell, et al., 1983). Initial approaches of ML focused on simplistic reasoning and rule-based systems that relied on predefined "if–then" rulesets created to imitate human decision-making and solve problems through logical inference rather than learning through data (Felgenbaum, 1977).

As computational power and data availability evolved throughout the years, it enabled statistical and data-driven methods to become more practical. Before the widespread use of big data, ML techniques were applied in areas such as handwriting recognition, medical diagnosis support and speech recognition (Mitchell, 2006). The methods used included neural networks, decision trees and support vector machines among others, due to their ability to identify patterns and generate predictions from historical data (Jain, et al., 2000).

During the early 2000s, ML research significantly expanded as businesses and researchers began integrating predictive models into Decision Support Systems and data mining. This period settled the foundation of modern analytical systems since it demonstrated that ML could improve automation, classification accuracy, and strategic decision-making across different fields (Power, 2008).

Today, organizations produce large volumes of structured and unstructured data with the use of online platforms, sensors, enterprise systems and cloud-based infrastructures. This has increased data availability and has grown the demand for intelligent analytical tools capable of interpreting data and supporting informed decision-making processes (Power, 2014). Since the competition between businesses and between institutions is escalating, the ability to efficiently analyze information has become a substantial strategic advantage.

ML techniques enable pattern recognition from historical data and prediction generation with limited human intervention. Over recent years, the advancements in computational power, data storage and artificial intelligence have significantly improved the practical application of ML models in real-world scenarios. Techniques such as classification, regression, and ensemble learning are being widely implemented (Ganaie, et al., 2022).

Despite their rapid development and adoption, ML technologies are a continuous research field due to the complexity, scale and dynamic nature of modern business environments. Digital transactions, customer interactions and social media platforms are a few examples of heterogeneous data generators that suggest the ML future to be long. Traditional analytical approaches are often unable to manage such data-intense environments or provide efficient clarifications into strategic decision-making (L'Heureux, et al., 2017). Consequently, modern ML

research focuses on improving predictive accuracy, scalability, interpretability and automation in order to support today's business intelligence systems (Sharda, et al., 2019).

ML systems are able to support business functions in customer relationship management, fraud detection, financial forecasting and personalized marketing by identifying patterns and generating predictions from historical data (Wamba, et al., 2017). Advancements in ML algorithms enable businesses to navigate beyond descriptive but toward predictive and prescriptive analytics in order to reduce uncertainties and control user trends (Bertsimas & Kallus, 2019).

The growing integration of AI into enterprise systems has created the demand for more transparent, adaptable and efficient ML methods that have to be capable of operating in real-time business environments. Regarding this field, academic research contributes towards the improvement of algorithm reliability, the reduction of model bias and the enhancement of automated explainability (Balasubramaniam, et al., 2023) that ML systems can integrate into decision-making processes. As an outcome, ML research provides both theoretical and practical value by supporting operational efficiency and strategic competitiveness across multiple industries.

1.2 Problem Statement and Research Gaps

Since large volumes of structured and unstructured data are generated daily, businesses, healthcare institutions and research environments are among some domains that increasingly rely on analytical systems (Mishra & Misra, 2017).

Although ML technologies provide organizations with the ability to obtain insights from such data, many existing ML systems remain complex and difficult to use for inexperienced users. The traditional ML workflows might often require substantial knowledge in data preprocessing, feature engineering and model selection and result interpretation among others (Abdel-Karim, et al., 2021).

These technical requirements can create barriers for researchers or students that lack specialized data science experience. The increasing scale and complexity of datasets is an additional challenging factor for the implementation of proper analytical approaches, that can require a more adaptable or automated approach (L'Heureux, et al., 2017).

Even though Automated Machine Learning (AutoML) systems have emerged for the simplification of model development and predictive analytics, many existing solutions are commercially expensive, computationally demanding or insufficiently interpretable for educational or research environments.

In many cases, AutoML platforms prioritize predictive performance which comes at the cost of limited transparency regarding model behavior. This lack of interpretability can reduce trust in the results and limit the practical integration of ML systems into the decision support process. The importance of explainability and user-centered ML systems capable of supporting understandable and reliable decision-making in real-time environments appears to be growing (Ezeji, et al., 2024).

Therefore, the research need for an accessible interactive system that automates predictive analytics while providing interpretable visual outputs appears to be essential. The ongoing necessity for accessible and automated analytical systems is addressed in this research together with machine learning techniques for decision support and the benefit of precise data-driven conclusions (Wang Aijie & Liang, 2023).

1.3 Aims and Objectives

This project aims to contribute to the fields of ML, pattern recognition, and visual analytics with the use of a developed interactive system that combines automated model generation with visual interpretation of results. The proposed system is able to support users in the identification of key patterns and predictive factors within datasets for the aid of data-driven decision-making.

The objectives of this project fall under design principles of the application interface and its implementation of an accessible ML-based decision support environment. In addition, the system's capability of automating key stages of predictive analytics such as data preview, configuration, model training, evaluation and result visualization aims to reduce the technical complexity of ML workflows by providing a structured step-by-step interface.

Moreover, the use of appropriate visual outputs and performance metrics aim to assist in explainability and understanding model behavior. This study intends to evaluate the effectiveness of the developed system in the identification of meaningful patterns and its strength in decision-making support. For this aim to be carried out, testing is conducted together with interpretation and visualization of results.

2 Literature Review

The literature review examines the theoretical and practical foundations, methodologies and existing research related to ML, predictive analytics, decision support systems as well as their application and ethical implications in their environments of integration.

Furthermore, this review considers how business support systems harness analytical models for speed and consistency in decision-making (Sharda, et al., 2015).

The aim of the literature review is to present and evaluate the strengths, limitations and practical uses of current ML applications and research in analytical and organizational contexts in order to establish an academic and technical background that supports the development of the proposed system.

2.1 Foundations of Machine Learning

ML has become an essential aspect of predictive analytics, for its ability to efficiently analyze large and complex datasets in order to identify patterns, forecast outcomes and support decision-making processes. The widespread availability of big data as supported by (Manyika, et al., 2011) has transformed modern economies, making data-driven strategies essential for innovation and competitive advantage.

2.1.1 Supervised and Unsupervised Learning

It is well established that supervised and unsupervised learning represent two of the most common and widely used ML paradigms in data analytics and predictive modelling. These approaches enable systems to identify patterns and relationships within datasets although there is a significant difference between them in terms of data structure, learning objectives and actual implementation (Pothos, et al., 2011).

Supervised learning relies on labelled datasets where input data is associated with target outputs that have been established. In such cases, the objective is to learn the relationship between variables for the generation of predictions or classifications to be possible (Love, 2002). Supervised learning techniques such as classification and regression are commonly used approaches (Rane, et al., 2024) in the prediction of categorical or continuous outcomes.

This learning approach is widely applied in areas that include fraud detection, medical diagnosis or customer behavior prediction.

In contrast, unsupervised learning operates on unlabeled data where predefined outputs do not take place. In such environments, direct predictions are not generated but instead unsupervised learning algorithms attempt to discover hidden structures, correlations or groupings within given data (Huang & Toyoizumi, 2016).

Applications of unsupervised learning include customer segmentation, anomaly detection and market basket analysis. The primary objective of unsupervised learning is exploratory analysis and pattern understanding through discovery rather than predictive accuracy (Jain, et al., 1999).

Examples of supervised learning application include the research of (Herland, et al., 2014) where the use of big data analytics and ML techniques are applied within Health Informatics for the extraction of significant findings from multi-level medical datasets. For medical diagnosis, supervised learning was applied through clinical studies that aim to map patient data to known medical conditions. In population and clinical informatics, supervised learning techniques appear to be relevant for the prediction of epidemic trends or in the identification of groups that would be at high risk.

To an equal extent, the paper of (Silva & Bernardino, 2022) presents an experimental study on the application and evaluation of both supervised and unsupervised learning within decision support systems. The objective of their study was the comparison of different algorithms in the performance of binary classification and clustering tasks. In addition, the study focused on whether ML can improve the accuracy of decision-making in real-life scenarios.

Supervised learning was implemented using labelled datasets where models like Logistic Regression, Support Vector Machine, Decision Tree, Naïve Bayes, K-Nearest Neighbors and Random Forest were trained and tested through performance metrics and in their ability to classify outcomes such as the presence of a disease or employee-related conditions.

Unsupervised learning was tested using clustering algorithms such as K-Means, BIRCH and Spectral Clustering that were applied to unlabeled data for the identification of hidden structures.

These techniques were also evaluated on the same datasets and compared with the supervised methods.

The final results demonstrated that supervised learning was the most efficient for classification and predictive tasks in decision support systems. This method provided more accurate and reliable results compared to unsupervised approaches, especially with the involvement of labelled data.

Following this notion, the provision of classification and regression techniques as options for the support of predictive analytics tasks through the identification of patterns, relationships and trends can be justified.

2.1.2 Supervised Learning Application in Predictive Tasks

Classification and regression are two of the most fundamental predictive techniques within supervised learning that are widely applied in data analysis and decision support systems. Classification is used when the target variable consists of categorical values while regression is applied when the target variable is continuous and numerical (Loh, 2011).

These predictive techniques contribute to automated decision-making and pattern recognition since they enable ML systems to learn relationships and generate predictions without specified programming.

The study of (Gadge & Karande, 2022) includes an evaluation of supervised learning algorithms and demonstrates that these predictive techniques are able to recognize patterns and classify complex data structures. With their use, decision-making can be supported in the areas of healthcare and human resource analytics. Additionally, the findings of the study showed that supervised learning algorithms achieved higher predictive performance and relationship learning capabilities compared to unsupervised approaches.

In the financial domain, these techniques are mostly used for their prediction capabilities. An example of such implementation is presented in the study of (Patel, et al., 2015) where the focus was drawn towards the prediction of the movement of stocks including their direction and the stock price indices within the Indian stock market.

The research evaluated supervised learning techniques in their ability to forecast the increase or decrease of stock prices based on historical data. Four different supervised learning models were implemented, an Artificial Neural Network (ANN), Support Vector Machine (SVM), Random Forest and Naive Bayes. The models were trained using financial data from the span of ten years.

The followed supervised learning workflow used technical indicators from stock trading data for predictive relationships to be learned between variables and future movements of the market. The results demonstrated that all supervised learning models showed improvement when trend deterministic data representation took place. The findings confirmed that supervised learning techniques were highly effective for financial prediction and relationship learning as well as market analysis within decision support environments.

Based on this research, both supervised and unsupervised learning appear to be robust techniques with supervised learning being widely implemented in modern modelling and decision support. Building upon these foundations, the role of predictive analytics, pattern mining and extraction techniques is explored.

2.2 Predictive Analytics and Pattern Extraction

2.2.1 The Predictive Analytics Process

As a branch of advanced analytics, predictive analytics focuses on the prediction of future events through the analysis of current and historical data. Through predictive models, organizations can estimate the likelihood of future outcomes and support the decision-making process that aims for specific future results. This analytical approach has increased substantially together with the evolution of big data technologies since businesses increasingly depend on analytical systems for proactive and customer related solutions (Kumar & Garg, 2018).

The predictive analytics process consists of several stages that contribute to the development of predictive models. Figure 1 visualizes the predictive analytics process.

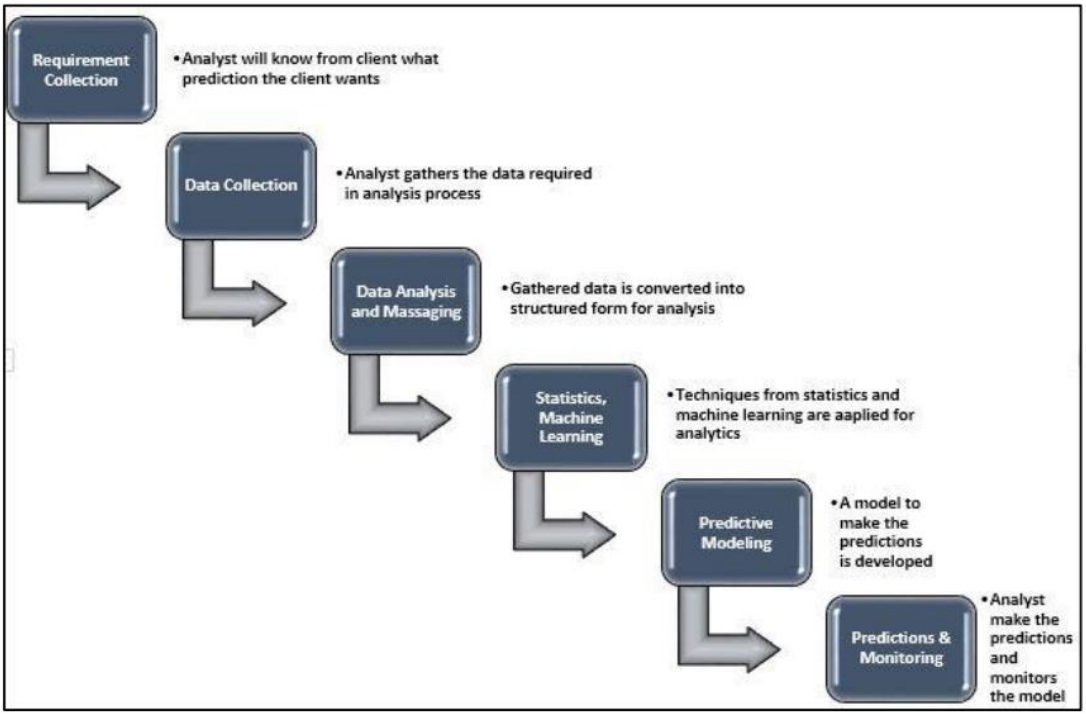


Figure 1 Predictive Analytics Process (Kumar & Garg, 2018)

Firstly, the requirements and prediction objectives need to be identified in order to determine the type of knowledge that the system should generate.

Next relevant data is collected from structured or unstructured sources and is prepared for analysis.

Data analysis and transformation are additional essential components since predictive performance depends heavily on the quality of the data.

Statistical machine learning techniques such as regression analysis, decision trees, support vector machines and artificial neural networks are some of the most widely applied tools in the "Statistics, Machine Learning" step for the development of predictive models.

After the models are trained and tested, they are deployed within corporate environments where they can generate predictions and provide analytical information.

Predictive analytics is applied in many organizational and analytical environments mainly for its growing dependence on data-driven decision-making. It is also supported by the development of user-friendly software which makes it accessible to not only analysts but managerial professionals as well (Kumar & Garg, 2018). For this reason, this analytical approach is integrated into businesses driven by the need for operation in competitive markets and effective response to altering economic conditions.

2.2.2 Pattern Extraction and Sequential Pattern Mining

As a feature generation step that aids in the improvement of downstream machine learning models, pattern extraction is directly linked with predictive analytics.

Based on the study of (Kadıoğlu, et al., 2023) pattern extraction refers to the process of identifying frequent, meaningful and structured patterns from large sequential datasets. It serves as a key component in pattern mining and instances that require discovery of regularities in data that cannot be immediately recognized. In this context, pattern extraction is conducted and presented using sequential pattern mining (SPM) techniques which allow users to define specific conditions while searching for data patterns.

The primary function of pattern extraction is the transformation of raw sequential data into structured, interpretable information that can be later used in analytics. This technique is used in real-world applications of digital behavior analysis, music streaming platforms and fashion e-commerce systems.

The results of pattern extraction can include the identification of correlations between sequences or feature representations that derive from relationships between use actions and outcomes.

Through pattern mining, patterns are associated with outcomes that enable automated feature creation for supervised learning models. These features can then be used to enhance predictive performance which makes predictions more accurate and interpretable (Kadıoğlu, et al., 2023).

A finding that can be extracted from this study is that pattern extraction not only supports descriptive analysis but can also reinforce predictive analytics by connecting raw sequential data with machine learning prediction models.

2.3 Decision Support Systems

2.3.1 The Evolution of Decision Support Systems

Decision support systems (DSS) emerged as an academic and organizational research field when researchers started exploring how computerized systems could support management and organizational decision-making processes. During the early period of DSS, influential studies established the foundations of DSS analysis, design and implementation within business environments. Among researchers, (Keen & Morton, 1978) had significant contributions towards the definition of DSS frameworks and different system types according to their functions and levels of data or model orientation.

DSS were developed for assistance in fields were managers and knowledge workers solved problems and analyzed data. Early systems ranged from data-oriented environments that provided access to stored information in model-driven systems that were capable of financial analysis (Power, 2008).

As the years went by, DSS expanded into organizational and executive support systems that could support recurring decisions or ad hoc analytical tasks.

Their use was driven by their ability to process data and apply analytical models for the generation of decision-oriented information. The study of (Power, 2008) covers the value of DSS applications for quantitative modelling, large database analysis and group decision-making support.

In later years, AI was integrated into DSS environments together with expert systems and communication technologies which allowed for more interactive systems capable of supporting increasingly complex organizational decisions (Power, 2008).

2.3.2 Design Methodologies and Implementation

Commonly used DSS include Business Intelligence (BI) platforms such as Microsoft Power BI or Tableau as well as enterprise systems like SAP Analytics Cloud. These systems are utilized due to their ability to handle large volumes of data and edit, present or handle it with the use of specific interfaces and interactive visualizations. Their adoption in today’s industry is initiated by the need for faster, more accurate and scalable decision-making in business environments (Balasubramanian, et al., 2024).

The design and implementation of DSS and ML-based predictive analytics solutions are usually results of a structured methodology that aims to ensure system effectiveness. The article of (Arinze, 1992) in the journal Systemic Practice and Action Research emphasizes the importance of requirements analysis in DSS development and the evolution of DSS methodologies.

DSS design is inherently more complex than traditional information systems so its requirements may be specialized. Thus, requirement analysis is significant in such cases where a guiding system is needed for the selection of an appropriate methodology.

A DSS methodology is used to provide guidance in theoretical approaches and the transition to specific technical solutions. This guidance offers a balance between theory and practical application.

Even though traditional methodologies that were established decades ago might still be widely used, they have been criticized for their lack of consistency, integration ability and documentation standards; since business plans try to adhere to the evolving market (Arnott & Pervan, 2014).

Structured methodologies on the other hand, adopt a different design suitable for well-defined problems that can ensure maintainability and better interactions within the subsystem.

Nevertheless, Many decision-making scenarios involve qualitative and subjective elements (Eldabi, et al., 2002) that lead to the development of "soft" methodologies. The "soft" design of implementation emphasizes human judgment, intuition and interaction. These approaches address the importance of adapting a methodology to a specific domain problem since the "soft" approach supports the idea of human capability and adaptability (Arinze, 1992).

A DSS implementation presented by (Goslar, et al., 1986) was evaluated through an experiment conducted over six days where different combinations of DSS availability, training, and data levels were applied to participant groups. Specifically, the implementation involved professionals from marketing and sales backgrounds in order to simulate realistic business decision-making conditions. The analysis included ANOVA and chi-square tests for the impact of decision-making performance to be assessed.

The results indicated that DSS training alone did not significantly improve decision-making performance. However, participants with no DSS support explored alternatives and different decision paths. This result indicated that DSS implementation alone was insufficient without proper user experience.

Paradigms of moder DSS are therefore justified when emphasizing user training and alignment with real-world scenarios, with the system capabilities they offer.

A recent implementation presented in the study of (Ali, et al., 2023) was based on systematic literature review approach aimed at the evaluation of Intelligent Decision Support Systems (IDSS) that integrate machine learning (ML) and multi-criteria decision-making (MCDM), The study was implemented by collecting and analyzing relevant research published between 2015 and early 2020.

In this implementation, various algorithms, frameworks and techniques combining ML, artificial intelligence and MCDM were examined across business, ICT and agriculture domains.

The results of this study DSS are widely adopted today for their effective tools on decision-making problems with multiple criteria. The study concluded that the integration of ML and MCDM can enhance DSS for more efficient solutions to be produced. Additionally, this implementation can support a more cost-effective and intelligent decision support framework (Ali, et al., 2023).

The study of (Silva & Bernardino, 2022) presents the implementation of a DSS through an evaluation of ML algorithms applied to real-world datasets of diabetes and human resources. The system was implemented using the Python-based Scikit-learn library and included supervised and unsupervised learning techniques for binary classifications.

Multiple algorithms were tested and compared based on performance metrics. The results showed that supervised learning methods like Random Forest achieved the highest performance due to their robustness to data noise. Unsupervised techniques alone performed worse but when used as a preprocessing step they improve the performance of supervised models in some cases.

Based on the implementation results, DSS seem to benefit significantly from the integration of ML since such an integration can enhance accuracy and efficiency. However, effectiveness depends upon data quality, model selection and a correct combination of analytical techniques (Silva & Bernardino, 2022). This suggests that human intervention is still required even in today's times.

2.3.3 Integration of Machine Learning in Decision Support

Classification models such as decision trees and support vector machines (SVM) are usually applied in tasks that regard customer segmentation (Rogić Sunčica & Ljiljana, 2021) or fraud detection. Regression models are used for forecasting data such as sales or product demand. Additionally, ensemble methods including random forests and gradient boosting are largely preferred due to their ability of prediction improvement (Callens, et al., 2020) by combining multiple models.

The decision-making processes can be directly supported by these techniques since their results can reduce uncertainty and ultimately improve the accuracy of the final conclusion. Predictive models allow businesses to identify influencing factors regarding their released products or the behavior of consumers. This step includes an evaluation of potential outcomes that a business judgement might introduce which is crucial before taking a final action and applying a framework in the real world. As a result, the ML models have to enable informed, consistent and timely decisions (Sharda, et al., 2015). The correct application of ML in this context can not only improve operational efficiency but also a competitive advantage by enabling proactive responses to emerging patterns and behaviors (Olalekan, 2019).

Using the capabilities of predictive models, DSS provide an environment where ML outputs can be used for evidence-based conclusions. DSS are interactive computer-based systems that integrate data, analytical models and interfaces to help decision makers such as analysts or product managers in finding solutions to semi-structured (Alexander N. Pavlov, et al., 2017) or unstructured problems (Courtney, 2001).

Modern DSS differ from traditional analytical tools since they can generate insights in real time by using incorporated ML techniques to process large volumes of data (Filip, 2020). The integration of ML within DSS enables automated analytical tasks such as pattern identification or prediction of potential outcomes. Such automation can reduce cognitive load on practitioners while giving them control over the final outcome. This supports the view of (Courtney, 2001) that

DSS does not replace human judgement but assists it by providing evidence-based recommendations, similar to claims by (Sharda, et al., 2015).

Today's intelligent DSS extends the capabilities of these systems by implementing artificial intelligence techniques that enable systems to learn from historical data and adapt accordingly (Filip, 2020). This capability is significantly valuable in today's business environments where rapid and accurate decision-making is a critical step to success.

The modernization of analytics supporting systems has set a new standard of DSS that automate processes of predictive analytics and pattern extraction.

This claim is supported by (Wen & Li, 2022) in their research where a systematic review of the integration between Spatial Decision Support Systems (SDSS) and AutoML is presented. The aim of it is to improve user adoption with the reduction of technical barriers in analytical systems. It is explained in the study that SDSS environments are not widely adopted due to limited user trust or lack of technical expertise among others.

To address these challenges, the study introduces AutoML as a solution that minimizes human intervention. A framework of SDSS with AutoML is proposed for the simplification of predictive analytics and the reduction of complexity in data processing and model development.

The study concludes that integrating AutoML into DSS can improve usability, reduce implementation costs and support more efficient data-driven decision-making.

2.4 Automation in Machine Learning and Decision Support

The automation of ML processes appears to have become an important component within modern decision support environments as supported by (Hutter, et al., 2019). Automation in Machine Learning and Decision Support focuses on the reduction of complex workflows from predictive analytics that can be generated based on the depth level of a conducted analysis and the data size.

This reduction of complexity takes place through the integration of intelligent systems capable of handling tasks such as data processing, model generation, evaluation and interpretation with limited human intervention. This integration in DSS environments allows for improved support for users with faster and more informed decisions as well as better predictive performance (Poloju, 2024).

2.4.1 Automated Machine Learning (AutoML)

AutoML is presented as a modern approach that lays within ML and focuses on the automation and optimization of predictive models. Its aim is to function successfully with minimal human intervention. AutoML is used for the simplification of complex ML workflows that would otherwise require advanced technical expertise (Wen & Li, 2022). This allows for faster development that can also reduce implementation time and operational costs.

The automated tasks inside an AutoML process can include data preprocessing, feature selection, algorithm selection, hyperparameter tuning, model generation and model evaluation (Filippou, et al., 2023).

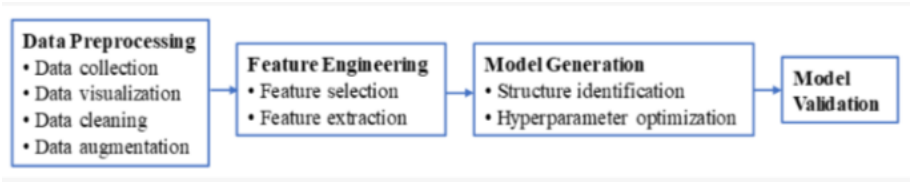


Figure 2 Pipeline of an AutoML approach (Filippou, et al., 2023)

Each of the presented stages involve additional subtasks that are conducted and integrated in a specific application.

A study focusing on the evaluation of AutoML tools is presented by (Romero, et al., 2022) where highly imbalanced medical datasets are used for the research purpose of comparing the predictive performance of different AutoML frameworks on healthcare prediction tasks. For this aim to be carried out, the researchers generated a large dataset from historical administrative medical claims which included disease-related codes that have been collected over multiple time periods.

The use of AutoML in this study is presented with the implementation of three different systems AutoSklearn, H2O and TPOT. These tools were trained on the same datasets for the prediction of disease outcomes. The AutoML frameworks automated major ML tasks including model generation, optimization and evaluation.

The performance of these tools was compared with the use of a baseline Random Forest model across model performance metrics. The benchmarking process included bootstrapping procedures and large validation sets on millions of healthcare records for reliable evaluation results.

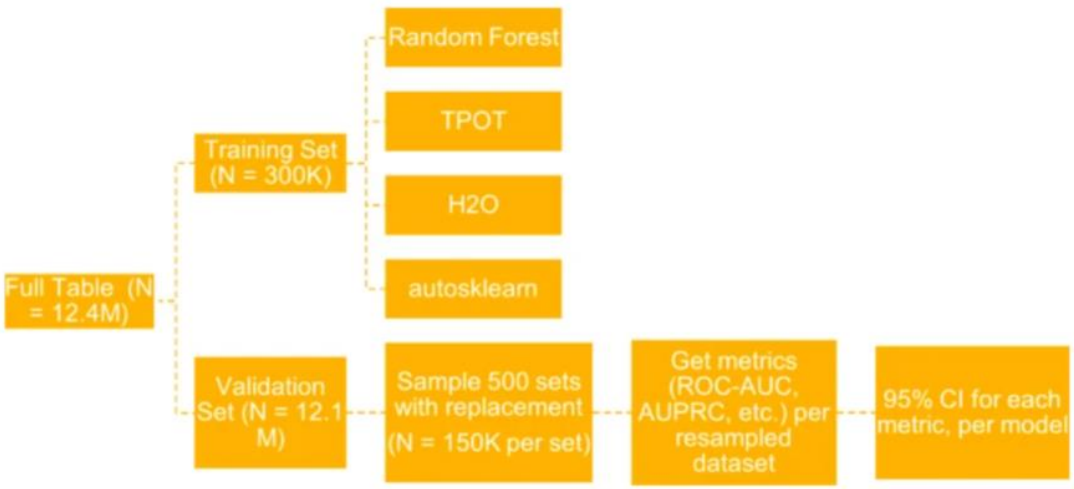


Figure 3 Flowchart of framework for benchmarking AutoML tools adapted from Gjisbers et al. As cited by (Romero, et al., 2022)

The above flowchart of the study depicts the AutoML frameworks and the Random Forest model trained on the same training set. The validation set appears to be resampled for the evaluation of model performance using ROC-AUC and AUPRC and 95% confidence intervals metrics.

The findings from this implementation showed that AutoML tools improved the predictive performance in comparison to the baseline model. Despite that, there was no particular AutoML framework that performed better than the others. It is concluded from the study that although AutoML can help predictive analytics and reduce the complexity of manual ML, healthcare data still presents challenges such as class imbalance or feature limitations.

Moreover, the study of (Zöller & Huber, 2021) presents both a conducted survey and a large-scale empirical evaluation of AutoML frameworks and hyperparameter optimization (HPO) algorithms.

As a primary objective, the study focuses on the investigation of AutoML and its effect on the reduction of the need for specialized ML expertise by enabling the automation in building ML pipelines without extensive knowledge. In addition, the paper covers AutoML techniques for the automation in different stages of an ML workflow.

The researchers evaluated a total of 14 frameworks including six AutoML frameworks and eight HPO approaches. The systems were tested on real-world datasets, and the study reviewed how these frameworks automated tasks such as preprocessing, feature selection, model selection, pipeline generation and hyperparameter tuning. Throughout the paper, mathematical formulations exist for the representation of the process of automatic ML pipeline synthesis.

Based on the results, most AutoML and HPO frameworks performed similarly to each other in terms of predictive accuracy. Simpler approaches such as random search did not get outperformed by more advanced methods.

A finding that emerged from this study is that individual dataset performance could vary depending on the framework used. In the end, the study concluded that AutoML significantly supports the automation of ML workflows and enables effective predictive models from a process that does not require extensive human expertise. Despite that, current human experts can still outperform automated systems in more advanced tasks.

To potentially address this limitation, the study of (Zimmer, et al., 2021) introduces Auto-PyTorch Tabular as an advanced AutoML framework designed for complex ML tasks that involve deep learning and neural architecture search. The study's focus is on the full automation of deep learning (AutoDL) that functions when neural network architectures and training hyperparameters are jointly optimized. As the researchers of the study aim for the simplification of deep learning implementation while maintaining strong predictive performance, tabular datasets are used from healthcare, finance, manufacturing, climate science and recommendation systems.

To support this research the study introduces LCBench as a benchmark dataset that contains learning curves from 2,000 configurations across 35 datasets. This benchmark was created for the systematic evaluation of multi-fidelity optimization approaches and to understand

how architectural and training hyperparameters can affect performance. The study's framework consists of a combination of advanced techniques including Bayesian optimization, ensemble learning and multi-fidelity optimization.

The choice of Auto-PyTorch appears to be strong, since it is claimed that this technique incorporates model selection strategies such as cross-validation, automated validation splitting and early stopping in order to improve generalization and prevent overfitting. The study presents a comparison of Auto-PyTorch Tabular with several established AutoML frameworks. Experimentation is conducted under equal runtime conditions and evaluated across multiple datasets.

The results appear to present Auto-PyTorch Tabular with the best overall performance and significantly better than its competitors. As a result, the researchers concluded that the combination of automated neural architecture search with ensemble learning and hyperparameter optimization can substantially improve AutoML performance for complex tasks.

In a related study by (Feurer, et al., 2022), the development and evaluation of advanced AutoML approaches through PoSH Auto-sklearn and Auto-sklearn 2.0 is presented for the improvement of automation and efficiency in ML pipeline generation. The focus of the researchers is drawn upon the reduction of complexity that exists in the designing phases of ML pieplines and the efficient performance of AutoML on large datasets under strict computational and time limitations.

The study introduces PoSH Auto-sklearn as an upgrade to the Auto-sklearn framework. The presented system leverages a portfolio-based meta-learning technique and a budget allocation strategy called Successive Halving (SH). This method is able to allocate computational resources to ML pipelines while reducing time spent on poorly performing configurations. By taking this action, the framework is able to improve optimization efficiency and increase the model selection speed.

The researchers automated the configuration process of the AutoML system itself, through algorithm selection techniques which led to the creation of Auto-sklearn 2.0 as presented in their study. For the evaluation of the framework, experiments using 208 meta-datasets for system construction and 39 benchmark datasets for performance evaluation were conducted. The study compared Auto-sklearn 2.0 with several AutoML frameworks including the original Auto-sklearn 1.0 across performance metrics such as multiclass log loss, binary classification AUC measures and ranking comparisons across datasets.

The results demonstrated that Auto-sklearn 2.0 was the most powerful since it achieved the best overall ranking among the evaluated frameworks. The system significantly improved predictive performance while it reduced optimization time. The research concluded that internal AutoML configuration together with other aspects like automated portfolio selection and budget allocation can substantially improve the robustness, scalability and efficiency of modern AutoML systems.

AutoML appears to contribute to efficient predictive analytics, improved decision-making and scalable deployment of intelligent systems. However, whether systems or processes are transparent enough is a question that can be further obscured, especially in more advanced deep

learning or similar system applications that literature itself might not be able to fully cover (Bjerring, et al., 2025).

2.4.2 Explainable AI in Machine Learning and Decision Support Systems

The study of (Zöller, et al., 2023) presents XAutoML as an open-source visual analytics tool made for the improvement of transparency and explainability of AutoML systems. It is explained in their study that even though AutoML frameworks are able to generate competitive end-to-end machine learning pipelines with almost no human intervention, struggles of trust to these systems appear often. This is caused by the lack of transparency and explanations regarding how pipelines and different procedures operate.

To address this issue, a requirements analysis is conducted and presented that involves 36 participants, including data scientists, domain experts and AutoML researchers with different levels of ML experience. The analysis findings present the developed XAutoML tool which combines interactive visualizations with explainable AI (XAI) techniques.

The study results showed a significant improvement in understanding of both AutoML processes and generated ML pipelines on participants who used XAutoML. On that ground, the researchers support that explainable and transparent AutoML environments can aid users to better validate automated decisions and increase their confidence in AutoML implementations.

According to (Kostopoulos, et al., 2024) the relationship between automation in ML and modern DSSs seems to be growing especially in the development of Explainable Decision Support Systems (XDSS).

The study elaborates on the increased integration of AI into DSS for the analysis of large data, pattern identification and predictive analytics. Many ML models however, operate as “black boxes” with increased complexity which means that even though high predictive accuracy can be achieved, the internal reasoning of such systems is often difficult to understand and interpret.

The following figure shows the trade-off of accuracy and explainability of some widely used AI models.

Automation within ML appears to contribute significantly to the improvement of DSS since it allows systems to process data automatically and generate predictive models for the support of analytical tasks with limited human intervention.

The study draws focus on how through automated explainability methods, DSS can become more transparent and understandable to end-users. Through this automation, manual analytical effort is reduced while processing, efficiency and decision-making are accelerated.

Explainable DSS are presented as an evolution step of this automated process by combining predictive machine learning capabilities with explainability techniques.

In their study, (Kostopoulos, et al., 2024) explain that XDSS are progressively applied in fields that include healthcare, manufacturing and education where understanding the reasoning behind predictions is critical. In such domains, explainable systems appear to be able for the creation of a

more informed, trustworthy and collaborative decision-based environment by improving user adoption in intelligent analytical technologies.

The paper of (Waa, et al., 2021) introduces Explainable Artificial Intelligence (XAI) as a growing research field that has emerged from the rapid evolution of AI. Similarly to (Kostopoulos, et al., 2024) the researchers support the idea that this field increases transparency and interpretability within machine learning models.

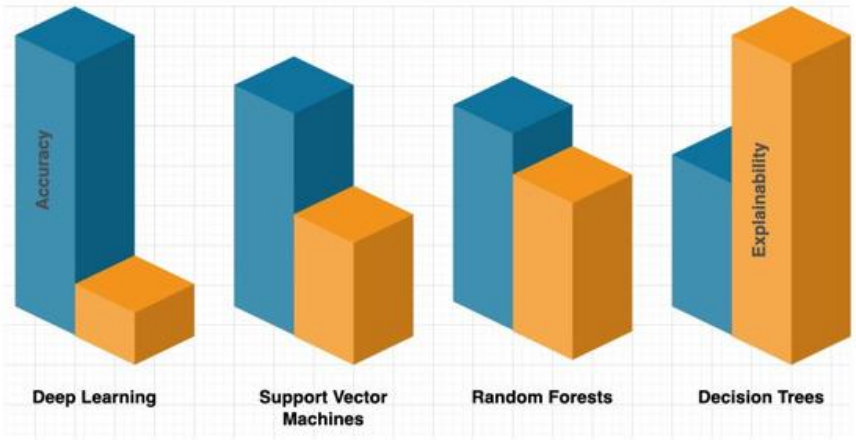


Figure 4 Accuracy vs. explainability trade-off for familiar AI models. (Kostopoulos, et al., 2024)

According to the study, many modern AI systems operate as "black boxes" and cannot be immediately interpreted (Waa, et al., 2021). The study presents an evaluation of two explainable approaches within AI-based decision support for self-management in cases of diabetes. The study includes a comparison of contrastive rule-based explanations and contrastive example-based explanations for the examination of their influence on user understanding and persuasiveness.

The evaluation investigates whether explanations helped users to better understand how the system generated recommendations catered to their case and whether these explanations aided in the effectiveness of decision-making.

The results demonstrated that rule-based explanations slightly improved understanding by identifying decisive factors behind recommendations. However, no explanation approach managed to significantly improve users' ability to predict system behavior. Despite this, both explanation methods increased the likelihood that users would follow the advice generated by the system.

Therefore, the study concluded that explainability alone does not necessarily improve decision quality especially when explanations fail to provide deeper reasoning behind system outputs.

An insight that can be extracted through this study is for effective explainability to take place within user-centered evaluations, more simplistic explanations or interpretable systems might need to be implemented.

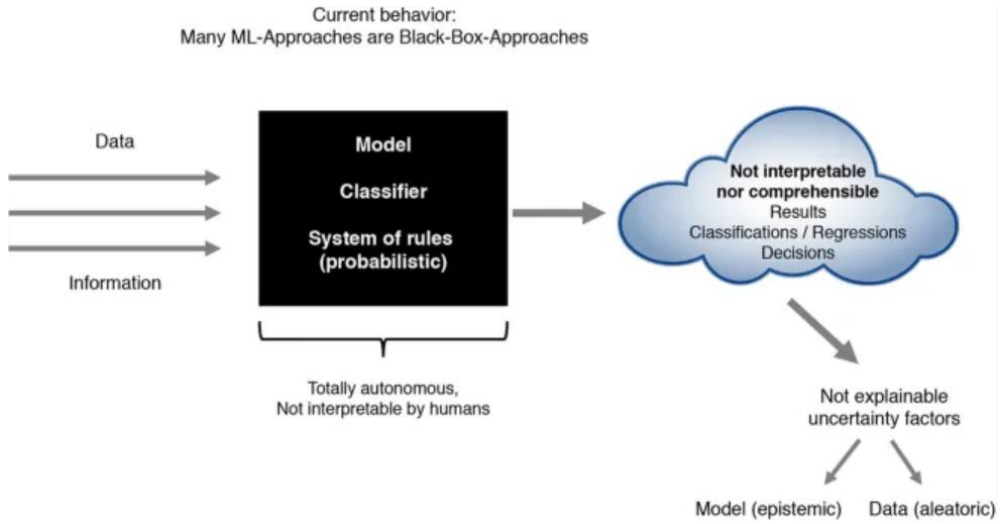


Figure 5 Present ML-approaches as black-box approaches. (Bruckert, et al., 2020)

The need for transparency and interpretability has driven the study of (Schwalbe & Finzel, 2023) into presenting a comprehensive overview of XAI. According to the study, the way in which advanced ML models operate can create challenges regarding trust, fairness, safety and legal compliance, especially in environments where AI systems are used to support decision-making.

The study conducts meta-analysis of existing XAI literature in order to develop a taxonomy of explainable methods. Instead of introducing a single XAI model, the paper organizes and compares various techniques according to their functionality and application.

It is explained that different XAI methods operate in unique ways depending on requirements of use cases. While some methods focus on local explanations by interpreting individual predictions, others provide global explainability by interpreting model behavior as a whole. Further distinctions are made between model-specific and model-agnostic methods, visual and rule-based explanations.

The study demonstrated that XAI has become fundamental within intelligent Decision Support Systems since it can improve trust and understanding of automated decisions. According to the study, it is implied that DSS environments are increasingly in need of explainable AI approaches that are tailored to specific users or operational contexts.

Emphasis is also put on how explainability should not be treated as universal but dependable on interactions between the explanation method, the domain of application and the needs of decision-makers.

As presented in the study of (Schwalbe & Finzel, 2023), the following figure shows a diagram about a framework for Comprehensible Artificial Intelligence.

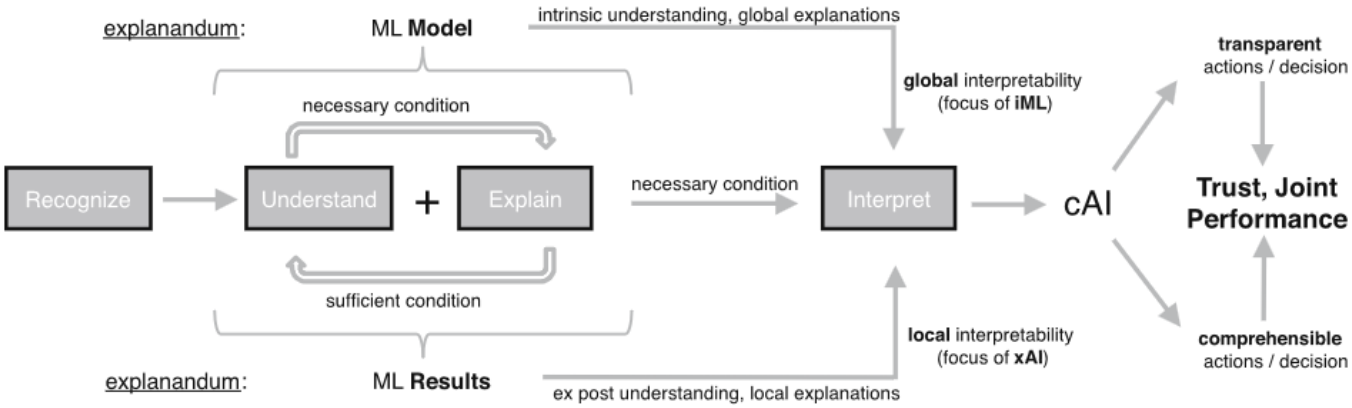


Figure 6 A framework for comprehensible artificial intelligence (Bruckert, et al., 2020) as cited by (Schwalbe & Finzel, 2023)

The framework shows the progression from data recognition to the understanding and explanation of ML behavior. “Understand” and “Explain” operate together, suggesting a mutual support between them. The presented processes interpret both the ML model (global interpretability, intrinsic understanding) and its output (local interpretability, post hoc explanations). The result depicts transparent and comprehensible AI decisions that encourage trust and the joint of human and AI performance.

By taking all into consideration, it is indicated that even though the automation in ML and DSS enhances efficiency and performance it can also introduce challenges that relate to transparency and interpretability. The integration of explainability mechanisms within AutoML-based systems seems to be able to bridge possible gaps by improving user understanding of automated decisions. This results in trust and human–AI collaboration in complex decision-making environments to be strengthened.

2.5 Considerations of Human–AI Integration in Decision Support Systems

Various studies cover the human-AI integration in systems and a good number of them support the existence of a human in that process. That however does not suggest the exclusion of automation.

The general application of ML has to be driven by an analyst that is able to solve possible challenges of technical complexity involved in model selection, data preprocessing and results. Thus, Automated systems, particularly intelligent DSS, become extremely important in the mitigation of human error while their automation can save valuable time (Wang Aijie & Liang, 2023).

Based on the study of (Storey, et al., 2024) that examines the evolution of DSS through the integration of AI and human interaction, systems based on collaboration are introduced called Human-AI Systems (HAIS). The study consists of a literature review based on the analysis of how these systems are designed and focuses on human interaction between human decision-making and AI.

The study covered the combination of human and system for the improvement of efficiency and decision quality in the past and today’s times. The conclusion supported that while in the past human involvement was central and decision-making was almost completely relied on human

judgement, today's standards follow the importance of human intervention but since the AI integration, an environment of role sharing between human and AI is promoted.

A similar concept of human intervention in DSS is covered in the study of (Lazaros, et al., 2026) that examines the concept of Human-in-the-Loop (HITL) as an approach to integrate human judgment into AI systems. This integration is mentioned to be particularly for high-stake decision-making situations where automation can be insufficient.

A systematic review on HITL methods provided in this study, focuses on how human intervention can direct decision-making. The study presents a development of a taxonomy that classifies HITL systems based on loop placement, level of interaction, and timing, offering a structured way to design and compare such systems.

Based on the study, HITL systems rely on commonly used techniques like active learning, reinforcement learning from human feedback, and explainability methods to enhance human and AI collaboration.

The findings indicate that the most effective approach for modern DSS is not full automation but a collaborative framework. However, scalability, cognitive load, and trust calibration indicate that system design has to be rigorous.

On the contrary, the study of (Sele & Chugunova, 2024) investigates how human intervention affects automated decision-making specifically in acceptance and accuracy of algorithmic recommendations. The researchers conducted an online experiment where participants were asked to predict student performance based on profile data. They received recommendations either from an algorithm or another human, and the experiment used a between-subject design to compare full delegation with an HITL condition where users could adjust recommendations.

The results of the study showed that most participants preferred algorithmic recommendations with increasing preference in instances where they could monitor and modify them. A key finding of the study was that human intervention reduced the overall decision accuracy. Participants were less likely to correct larger errors and were not effective in improving predictions.

As a result, the study concluded that HITL systems may increase trust, engagement, and acceptance but they can also reduce accuracy of results.

To a similar extent, a study that examines human intervention in automated decision-making presented by (Agudo, et al., 2024), evaluates how HITL systems in high-stakes contexts improve decision accuracy and whether they introduce biases.

For that investigation, the researchers conducted two experiments that judged participants on the guilt of defendants in simulated legal scenarios. The experiments adjusted the timing of algorithmic support to prove AI recommendations either before or after participants made their decisions.

The results showed that human decisions were strongly influenced by algorithmic input especially when it was presented before participant judgement. In such cases, incorrect AI recommendations

reduced decision accuracy. Participants that made their own decisions however, improved their accuracy and reliance on incorrect AI suggestions was decreased.

Based on these results, it can be claimed that even though human involvement is intended for security, it does not always improve outcomes. A poorly designed interaction can reduce effectiveness due to influence. Thus, human-AI collaboration is a sensitive structure.

2.6 Review Summary

The literature review established theoretical and practical foundations that prove to be necessary for the development of modern ML-driven systems. The importance of AutoML in this domain appears to be ever growing since the reduction of complexity in ML workflows through the automation of tasks including preprocessing, model selection, hyperparameter optimization and evaluation appear to be crucial. The presented implementations and benchmarking studies covered different AutoML frameworks that were able to achieve competitive predictive performance while reducing the need for extensive technical expertise.

Research on Explainable AI and human–AI collaboration further demonstrated that a careful implementation focused on interaction mechanisms is essential for trustworthy and effective DSS environments.

Collectively, the findings included in the literature review provide both the academic justification and methodological foundation for the proposed system whose design and implementation are presented in the following chapter.

3 DataPilot: Design, Methodology and Evaluation of the Proposed System

By expanding the theoretical foundations covered, this section moves toward the implementation of previous identified concepts in order to present and evaluate the proposed system for the support of decision-making.

Initially, this section introduces an existing industry architecture and methodology that relates to AutoML and decision support environments. The use of the following industry introduction aims to use a real implementation for reference purposes as a technical basis that informs the established development approach.

Following this, the proposed methodology of the system is presented including the workflow and additional procedures containing the integration of predictive and automated analytical components.

The focus is then drawn towards the system design and structure of key functionalities. In the end, the conducted testing and result evaluation is presented in order to assess the effectiveness and practical performance of the proposed implementation.

3.1 Reference Architecture

The study of (Wang, et al., 2021) presents an Automated Machine Learning (AutoML) system designed for the support and simplification of the data science lifecycle through workflow automation. The system is called AutoDS and was created for the aim of reducing manual and time-consuming tasks that are required in data science projects. Such tasks involve data preprocessing, model selection, configuration, and training.

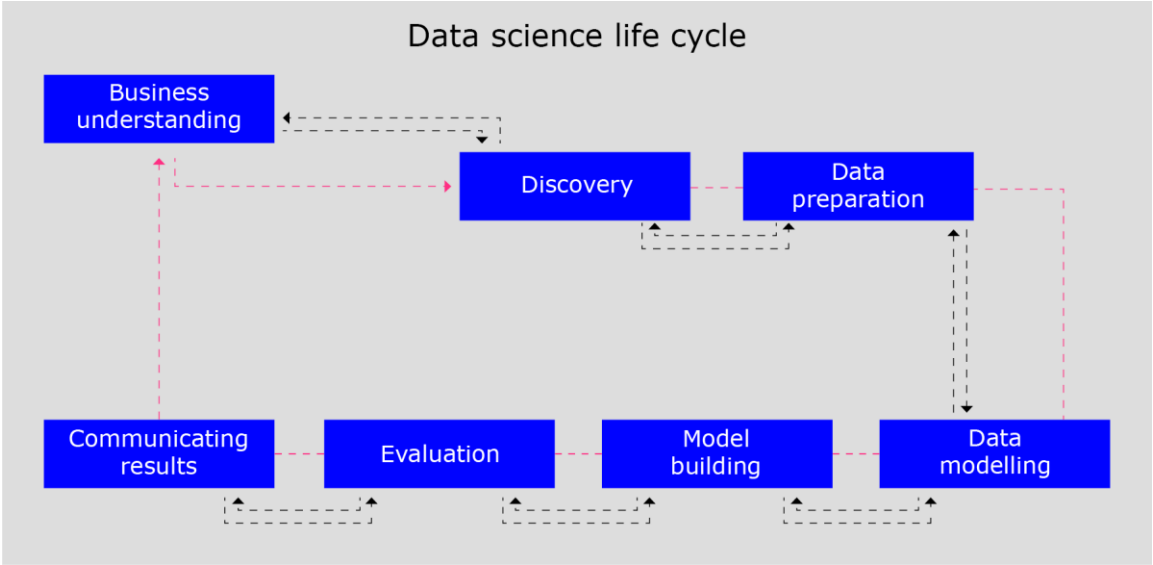


Figure 7 An example of the data science life cycle (FutureLearn, n.d.).

AutoDS is proposed as a human-centered AI system that combines automation and collaborative user interaction rather than being fully automated and independent of human intervention. The system is claimed to have been developed in response to the increase of complexity of ML workflows and the need for collaborative AI systems. Human–Computer Interaction (HCI) is particularly emphasized in this paper since transparency and user awareness are claimed to be significant in such processes.

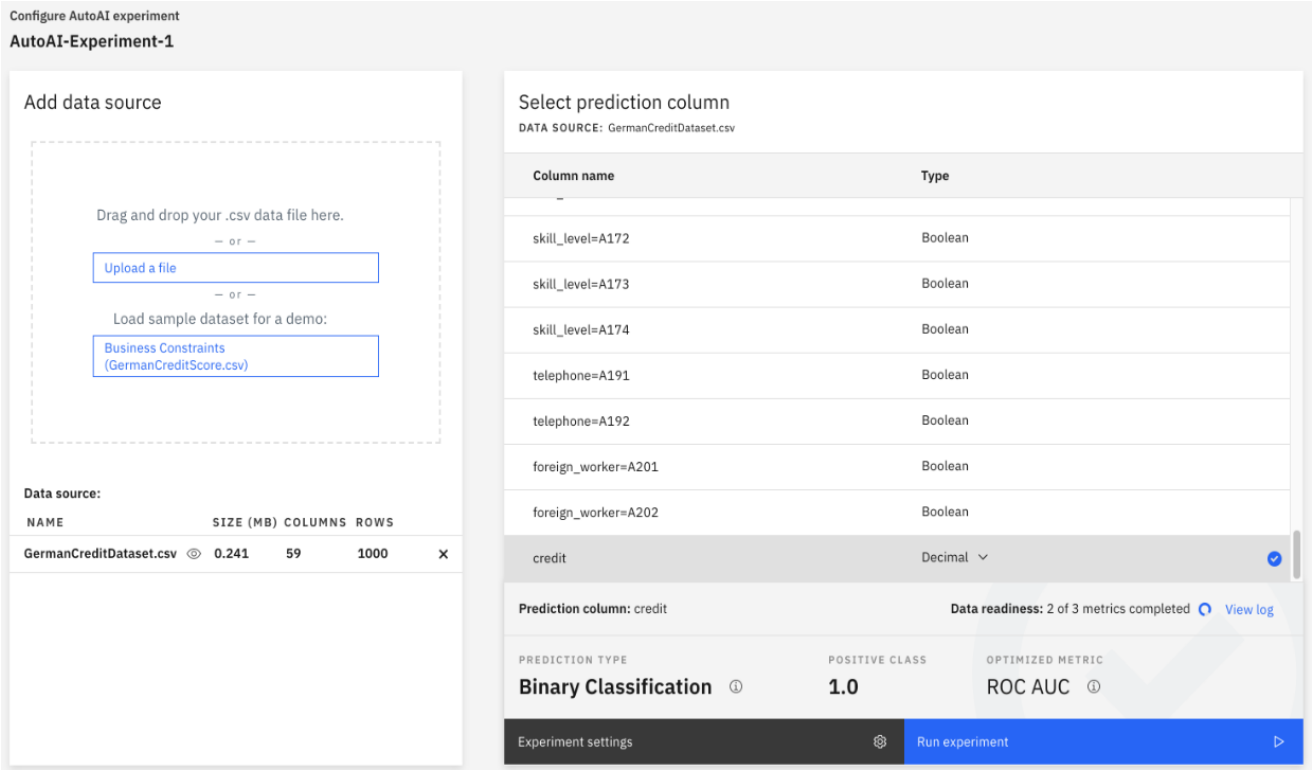


Figure 8 Configuration UI Screen (Wang, et al., 2021)

The system functions through an automated workflow that includes an initial upload of a dataset into the platform. The system then automatically suggests appropriate ML configurations, selects algorithms and trains candidate models. User interaction exists through a graphical interface that consists of a configuration screen and a results interface that provides a visualization of the AutoML process which displays tested algorithms, engineering and optimization steps.

The experimental findings showed that users working with AutoDS produced high quality models. However, the confidence of users appeared to have decreased in the automatically generated models.

User interaction appears to be a crucial factor in systems that introduce automation of choices inside the data science life cycle. Interactions within such systems are able to foster a collaborative environment that can ensure better analytical results and increase user trust in the system.

Using this approach of collaboration and system support on user decision making the following section presents the methodology and workflow of the proposed system influenced by the researched industry work.

3.2 Proposed System Design

In this section, the system design and functionality are presented using relevant diagrams regarding the workflow of the system and how each file within the program interacts for the final result to be carried out.

This section aims to provide additional transparency to the system's design so the use or testing of the system can be conducted with greater confidence. Below, the flowchart of the system is presented.

The diagram illustrates the workflow of the proposed system towards guiding a complete predictive modeling process.

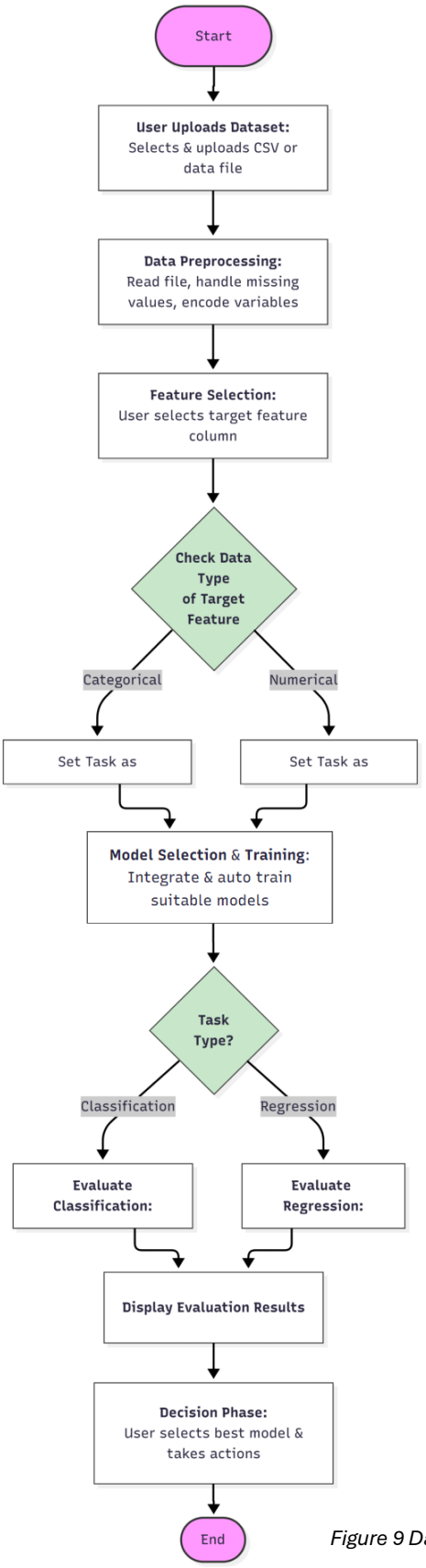


Figure 9 DataPilot Flowchart

In the beginning, the program starts with the dataset upload phase where users submit a CSV file containing the data to be analyzed. The second step includes data pre-processing which involves the processing of missing values as well as categorical features to make sure that the provided data is correctly prepared for analysis and model training. After these phases, users can choose the target feature of their data analysis task which would be the variable of prediction through ML modeling.

The program then detects the data type of the selected variable and makes the decision whether the task should be treated as a classification problem in case the target variable is categorical or as a regression problem in case the variable is numeric. Based on this decision, the ML modeling process can be further configured according to the selected problem category.

Next, the system enters the stage of model building, evaluation and visualization where the ML algorithms are trained for the selected tasks using the provided datasets and model evaluation methods including proper visualization techniques. In the case of classification, evaluation methods such as confusion matrices are used while in the case of regression, scatter plots.

Finally, the evaluation results enable a comparison between models that can be used in the decision phase where users select the most suitable model and determine subsequent actions based on the results and behavior of the models and data.

The following component diagram illustrates the software architecture of the system and presents the reasoning behind the system's layout and coordination on each step of its process.

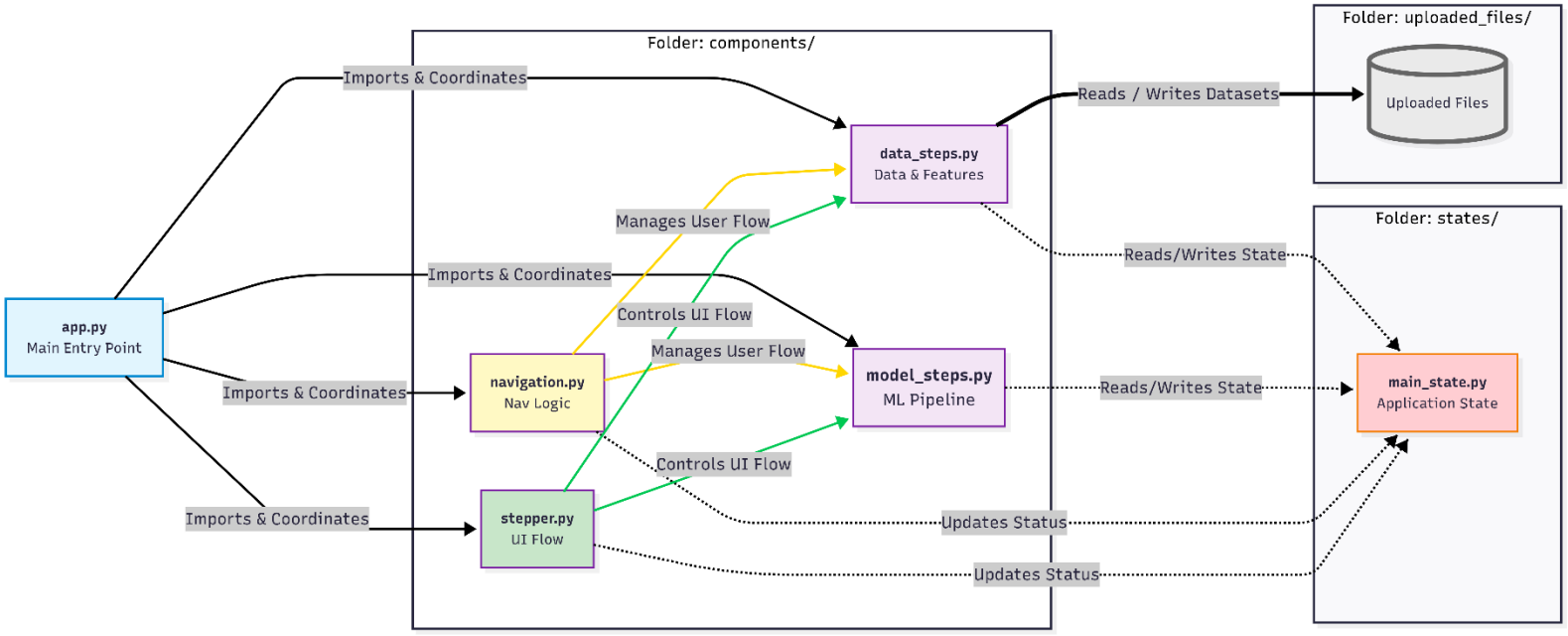


Figure 10 DataPilot Component Diagram

The system architecture revolves around *app* which functions as the primary component and application entry point. This component imports and coordinates all subcomponents including the initialization of the application environment and the connection of interface actions or routes to the appropriate processing functions.

Within the *components* directory, the system is divided into different modules. The *data_steps* is used for the management of the data lifecycle since it reads datasets from *uploaded_files* parses file streams and preprocessing data. In addition, this component interacts with the *main_state* file for the retrieval or storing of data during the system's execution.

The *model_steps* component includes the ML pipeline which consists of the coordination of model training, hyperparameter configuration, evaluation and prediction generation. The component exchanges information with *main_state*.

The *navigation* and *stepper* files are used for the workflow of the interface. These components manage the navigation between the stages of analysis so each analytical step can be structured in a more understandable way for users.

The *main_state* component is located in its own *states* directory and works as the centralized state management component. It enables the collaboration of each component and reduces direct dependencies between the files.

3.3 System Implementation Using Reflex

In this section, the primary tool of the system is introduced together with relevant code sections of importance with the aim of transitioning from theoretical information to practical presentation.

Reflex is the main tool used for the implementation of DataPilot. The selection of it is justified because of its structure as an open-source Python framework. Reflex is generally used for the development of full-stack web applications through a single codebase (reflex.dev, 2026). The use of a full Python environment supports the idea of smoother integration and functionality between data analysis (Nagpal & Gabrani, 2019) and coherent web applications. The Reflex framework allows for the creation of both frontend and backend entirely in Python without the requirement of separate web development frameworks (reflex.dev, 2026).

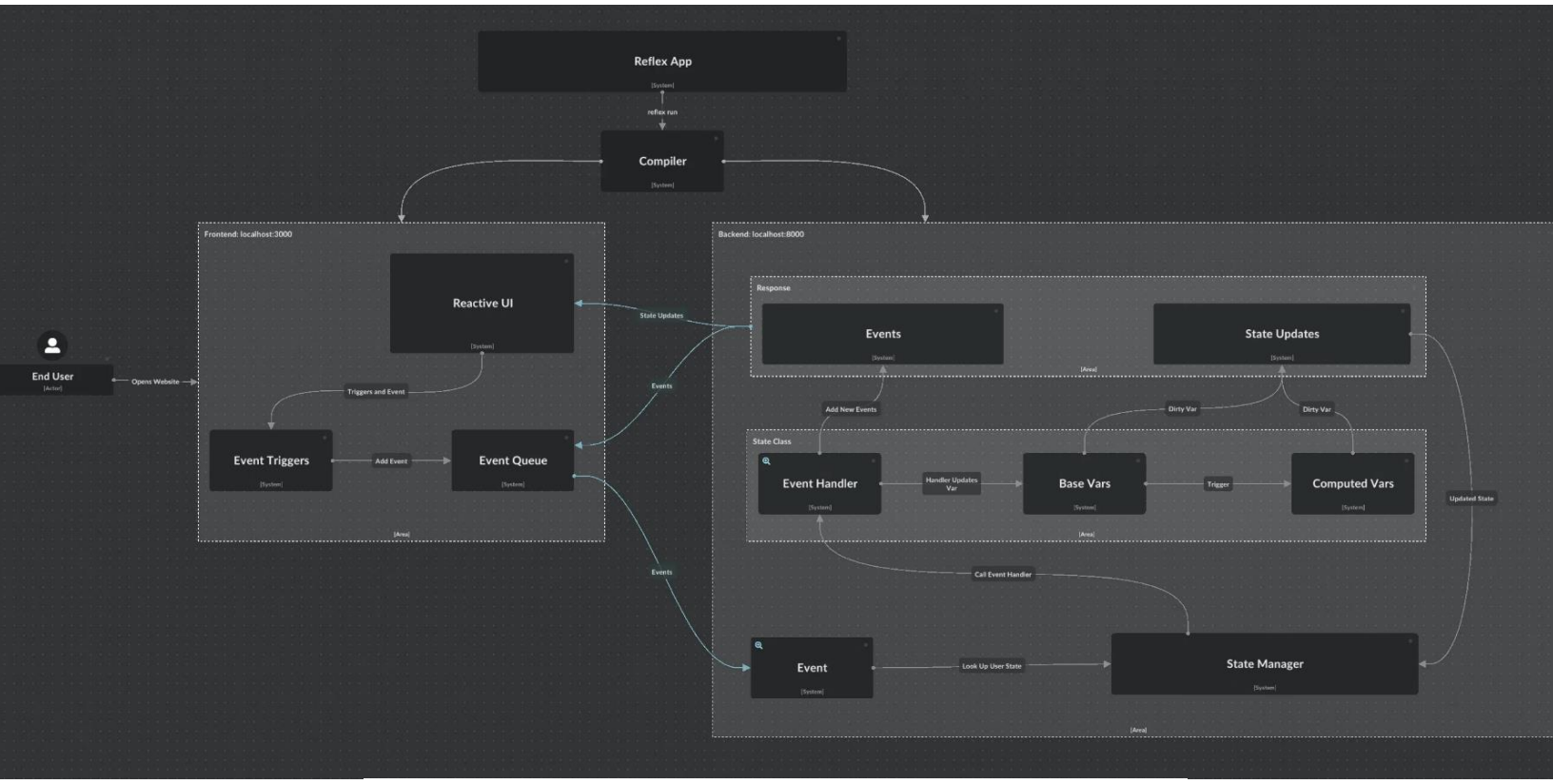


Figure 11 A detailed overview of how a Reflex app works (reflex.dev, 2026)

It functions by compiling the frontend into a React and Next.js web application while the backend operates through a FastAPI server. The frontend handles the user interface and transmits user-generated events, whereas the backend manages the application logic, state handling and data processing.

Application states within Reflex are defined by Python classes where variables represent dynamic data. Whenever users perform an action such as submitting input or pressing a button the event is sent into the backend and processed by the corresponding event handler and then reflected back to the interface.

For the development of DataPilot, Reflex was selected for its ability of faster implementation of interactive applications while providing efficient communication between interface and backend components. The implementation of a full Python environment can ensure the minimization of potential compatibility or scalability issues (reflex.dev, 2026).

3.3.1 Code Parts of Implementation

Important code figures are provided in this section for a better understanding of the system's internal functionalities. The following code presents how the background task of ML model training takes place without intervening with the main application.

```python
#MODEL TRAINING PIPELINE (RUNS IN BACKGROUND)
@rx.event(background=True)
async def train_models(self):

    #INITIALIZE TRAINING STATE
    async with self:
        self.is_training = True
        self.training_progress = 0
        self.training_status_msg = "Preprocessing data..."

        #CLEAR PREVIOUS RESULTS
        self.model_results = []
        self.feature_importance_plot = ""
        self.model_comparison_plot = ""
        self.confusion_matrix_plots = []
        self.pred_vs_actual_plots = []
        self.correlation_heatmap = ""
        self.best_model_name = ""
        self.best_model_score = ""
        self.show_data_validity_warning = False
```

Figure 12 Backround model training state initialization

The *@rx.event(background=True)* decorator makes the training run in the background which allows for the UI to stay responsive for users. The training state is initialized inside *train_models* with different flags being set like *is_training = True*, the reset of training progress and a status message. The second block clears older outputs from previous trainings.

The progress display of the training is provided in the following code snippet.

```
#ACTIVE STATE - TRAINING PROGRESS VISUALIZATION + STATUS FEEDBACK
rx.el.div(
    rx.el.div(
        rx.el.span(
            AppState.training_status_msg,
            class_name="text-sm font-medium text-gray-700",
        ),
        rx.el.span(
            f"{AppState.training_progress}%",
            class_name="text-sm font-bold text-gray-900",
        ),
        class_name="flex justify-between items-center mb-2",
    ),

    #PROGRESS BAR REFLECTS BACKEND TRAINING STATE IN REAL TIME
    rx.el.div(
        rx.el.div(
            class_name="h-3 bg-gray-900 rounded-full transition-all duration-500
            style={"width": f"{AppState.training_progress}%"},
        ),
        class_name="w-full h-3 bg-gray-100 rounded-full overflow-hidden shadow-i
    ),
```

Figure 13 The training progress layout and display

The training progress is displayed in real time with the outer rx.el.div() containers organizing the layout. In the first block, the current training status lies as AppState.training_status_msg and the percentage of the progressed training as AppState.training_progress%. These messages update while the backend runs. In the second block, the progress bar is bult. As the training progress increases, the bar updates accordingly.

The following part of code is of the target selection and the logic behind the ML task type that the system selects based on the target data type.

```
#TARGET COLUMN SELECTION AND TASK TYPE INFERENCE
@rx.event
def set_target_column(self, col: str):

    self.target_column = col

    #DETERMINE TASK TYPE BASED ON DATA CHARACTERISTICS
    if col and col in self.columns:

        info = next(
            (item for item in self.column_info if item["name"] == col), None
        )

        if info:
            unique_vals = int(info["unique"])
            dtype = info["dtype"]

            #HEURISTIC: SMALL UNIQUE VALUES OR CATEGORICAL CLASSIFICATION
            if dtype == "object" or dtype == "bool" or unique_vals < 20:
                self.task_type = "Classification"
            else:
                self.task_type = "Regression"

            self.target_stats = info
```

Figure 14 Target selection and task type inference

The *@rx.event* decorator makes *set_target_column* respond to actions taken by the user. Such an action in this case is the selection of a column from the UI. After users select a column, it gets

stored inside *self.target_column*. Then a check takes place to ensure the existence of the column and its data is extracted from *self.column_info*. Based on the column's data type (*dtype*) and number of unique values (*unique_vals*), categorical, boolean or low-unique-value columns are labeled as the classification task type while continuous numeric columns as regression task type. In the end, the statistics of the selected column are stored inside *self.target_stats*.

Next, the part of code that regards the preprocessing of data is presented.

```
try:
    #LOAD DATA FROM STORED BYTES
    df = pd.read_csv(io.BytesIO(self._df_bytes))

    #REMOVE COLUMNS WITH TOO MANY MISSING VALUES
    limit = len(df) * 0.5
    cols_to_keep = df.columns[df.isnull().sum() <= limit]

    #ENSURE TARGET COLUMN IS NOT DROPPED
    if self.target_column not in cols_to_keep:
        cols_to_keep = cols_to_keep.append(pd.Index([self.target_column]))

    df = df[cols_to_keep].copy()

    #MISSING VALUE HANDLING (NUMERIC → MEDIAN, CATEGORICAL → MODE)
    for col in df.columns:
        if df[col].isnull().any():
            if pd.api.types.is_numeric_dtype(df[col]):
                df[col] = df[col].fillna(df[col].median())
            else:
                mode_val = df[col].mode()
                df[col] = df[col].fillna(
                    mode_val[0] if not mode_val.empty else "Unknown"
                )

    #ENCODE CATEGORICAL VARIABLES USING LABEL ENCODER
    le_dict = {}
    for col in df.columns:
        if not pd.api.types.is_numeric_dtype(df[col]):
            le = LabelEncoder()
            df[col] = le.fit_transform(df[col].astype(str))
            le_dict[col] = le
```

Figure 15 The preprocessing steps

For the preprocessing of a selected dataset to take place, the data is loaded from stored byte data into a dataframe. Then the columns with 50% or more missing values are removed while the system ensures that the target variable is preserved. The handling of missing values includes the median fill on numeric columns and the most common value or Unknown if no mode exists, for categorical columns. In the end, categorical variables are converted into numerical values with LabelEncoder. Each encoder is stored in a dictionary for future use like the decoding of predictions.

For further information of model selection after correct preparation of data takes place, the following part of code presents the selection of appropriate models based on task type.

```
            #MODEL SELECTION BASED ON TASK TYPE
if self.task_type == "Classification":
    models = {
        "Logistic Regression": LogisticRegression(max_iter=1000),
        "Random Forest": RandomForestClassifier(random_state=42),
        "Gradient Boosting": GradientBoostingClassifier(random_state=42),
        "Decision Tree": DecisionTreeClassifier(random_state=42),
        "K-Nearest Neighbors": KNeighborsClassifier(),
    }

    #METRICS FOR CLASSIFICATION TASKS
    primary_metric_name = "Accuracy"
    secondary_metric_name = "F1 Score"

else:
    models = {
        "Linear Regression": LinearRegression(),
        "Ridge Regression": Ridge(),
        "Random Forest Regressor": RandomForestRegressor(random_state=42),
        "Gradient Boosting Regressor": GradientBoostingRegressor(random_state=42),
        "Decision Tree Regressor": DecisionTreeRegressor(random_state=42),
    }

    #METRICS FOR REGRESSION TASKS
    primary_metric_name = "R² Score"
    secondary_metric_name = "RMSE"
```

Figure 16 Model selection based on task type

The ML models and evaluation metrics are selected based on the task type. If the task type is set to *Classification* a dictionary of classification algorithms is created. The algorithms of classification include Logistic Regression, Random Forest, Gradient Boosting, Decision Tree and K-Nearest Neighbors. These algorithms have been selected for their widespread use on multiple domains as well as their easier interpretability, computational efficiency and simplicity of implementation (Alshammari, et al., 2026).

If the task is not classification, the case of regression takes place. In that case, regression models particularly Linear Regression, Ridge Regression, Random Forest Regressor, Gradient Boosting Regressor and Decision Trees are selected. The selection of such regression algorithms is supported by the comparative study of (Fernández-Delgado, et al., 2014) which evaluated 179 classifiers across 121 datasets were Random Forest, boosting methods, decision trees and related techniques appeared to be consistent in strong predictive performance and robustness in different domains.

In both cases, the appropriate evaluation metrics are initialized while a secondary metric exists for purposes of future work on the system.

3.4 User Interface

This section presents the UI of the system in parts structured based on each step of the analytical workflow that the system possesses. This section is essential before the presentation of the tests conducted, given that a clear understanding of the system can aid in more explainable testing and evaluation discussions.

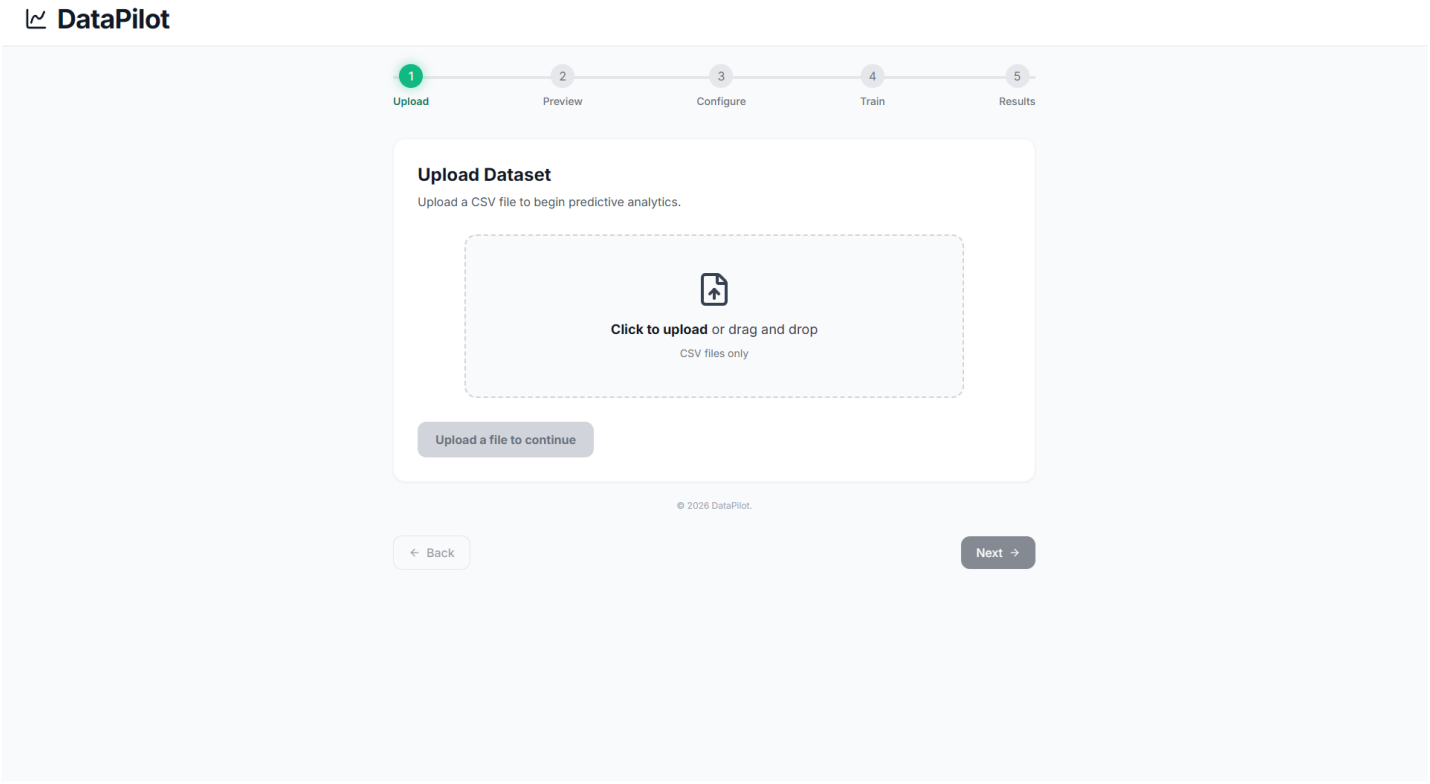


Figure 17 Initial upload stage interface

The above figure presents the initial upload page of the system. For this example, a dataset regarding Walmart sales is uploaded which will later be used for testing.

After the dataset is uploaded, the following page can be accessed that provides a detailed preview of the dataset.

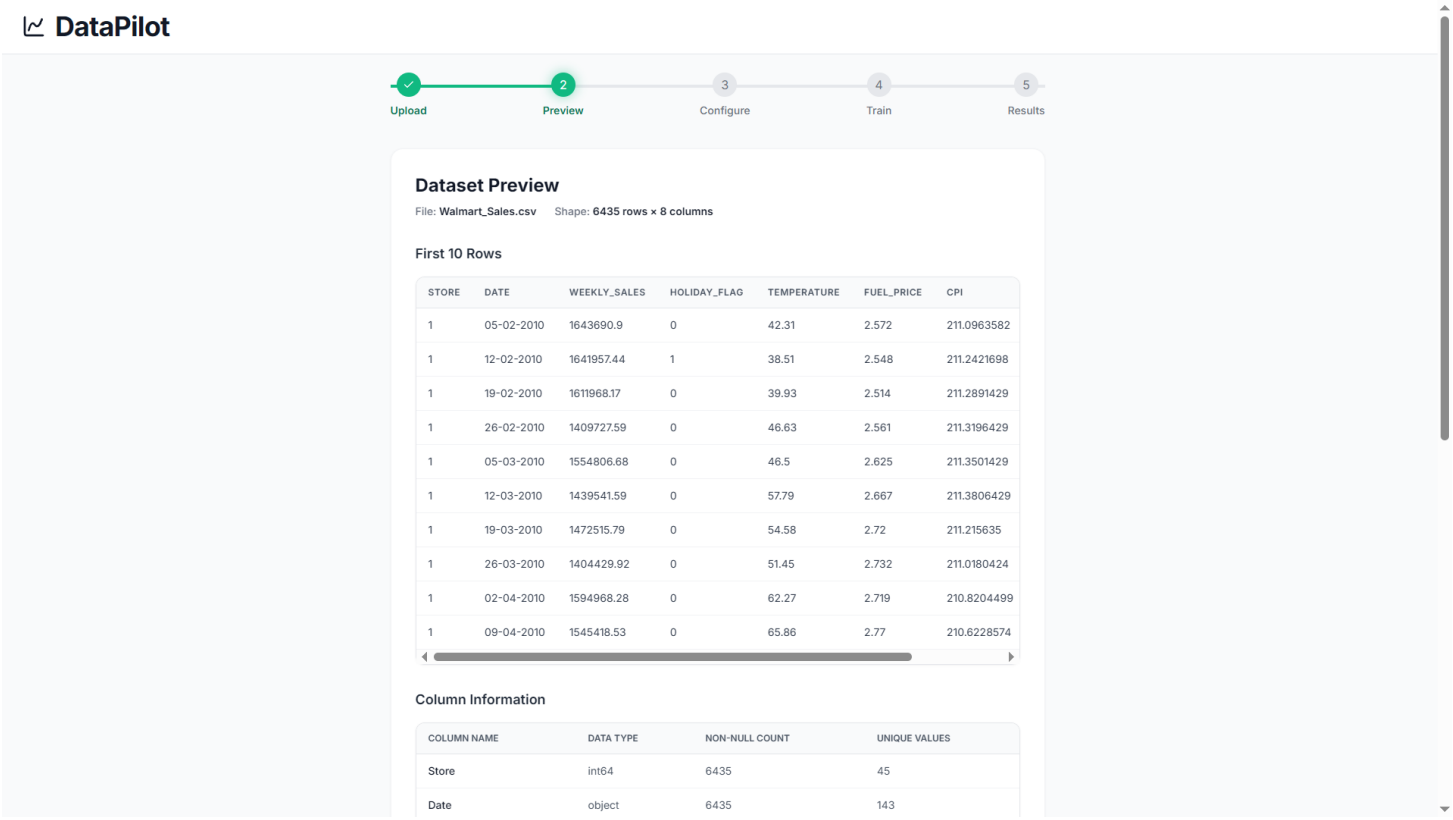


Figure 18 Uploaded dataset preview interface

After users are satisfied with the dataset information, they are able to move on to the target configuration and training split interface.

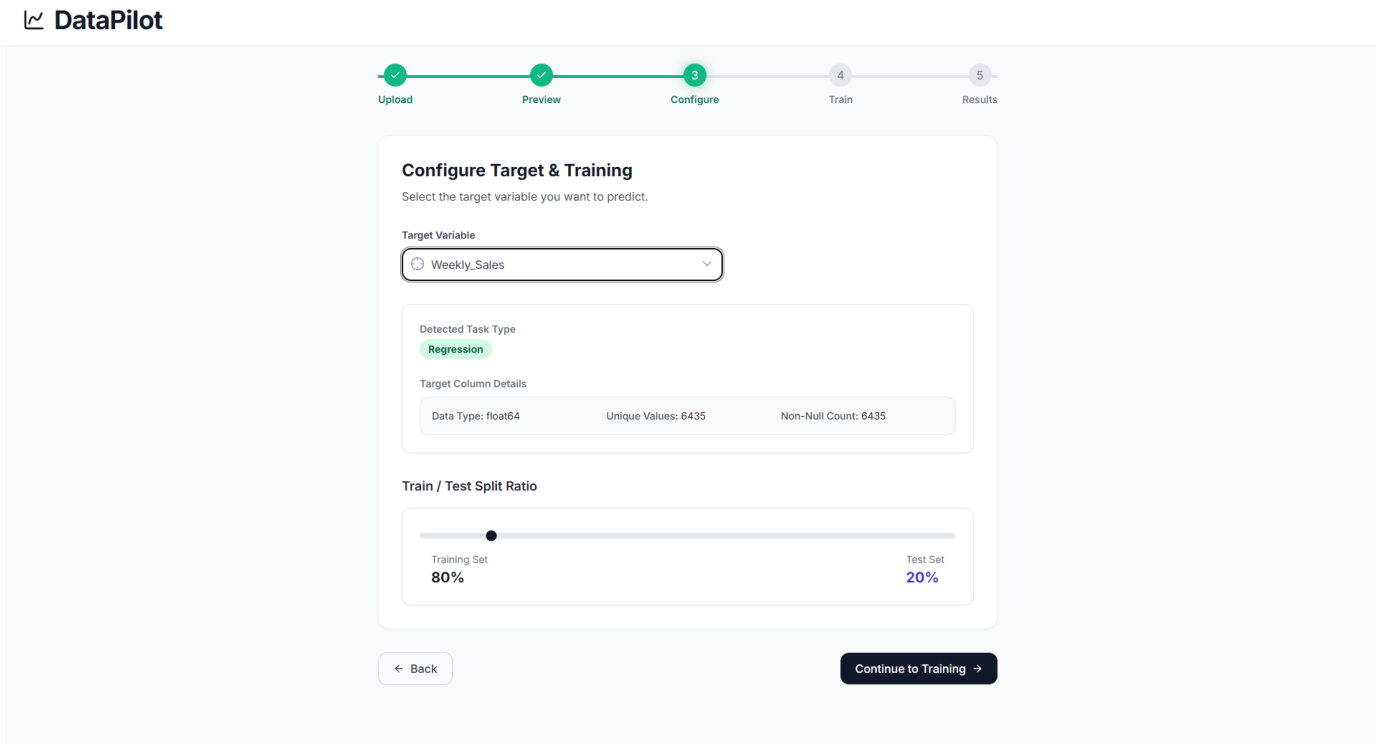


Figure 19 Analysis target and training split configuration interface

In this case, the weekly sales of the dataset can be analyzed with the Regression task type. Additionally, details about the target column are provided as well as the *Train/Test* split which works with a dragging bar in increments or decrements of 5 depending on the direction.

After the training and testing split is adjusted as well as the target variable, users can proceed to the model training interface. There the 5 defined models are trained whenever users press on the dedicated button. Before model training takes place, users can return to the previous interface by pressing the *Back* button. By pressing the *DataPilot* button at the top, they can return to the initial uploading interface at all times.

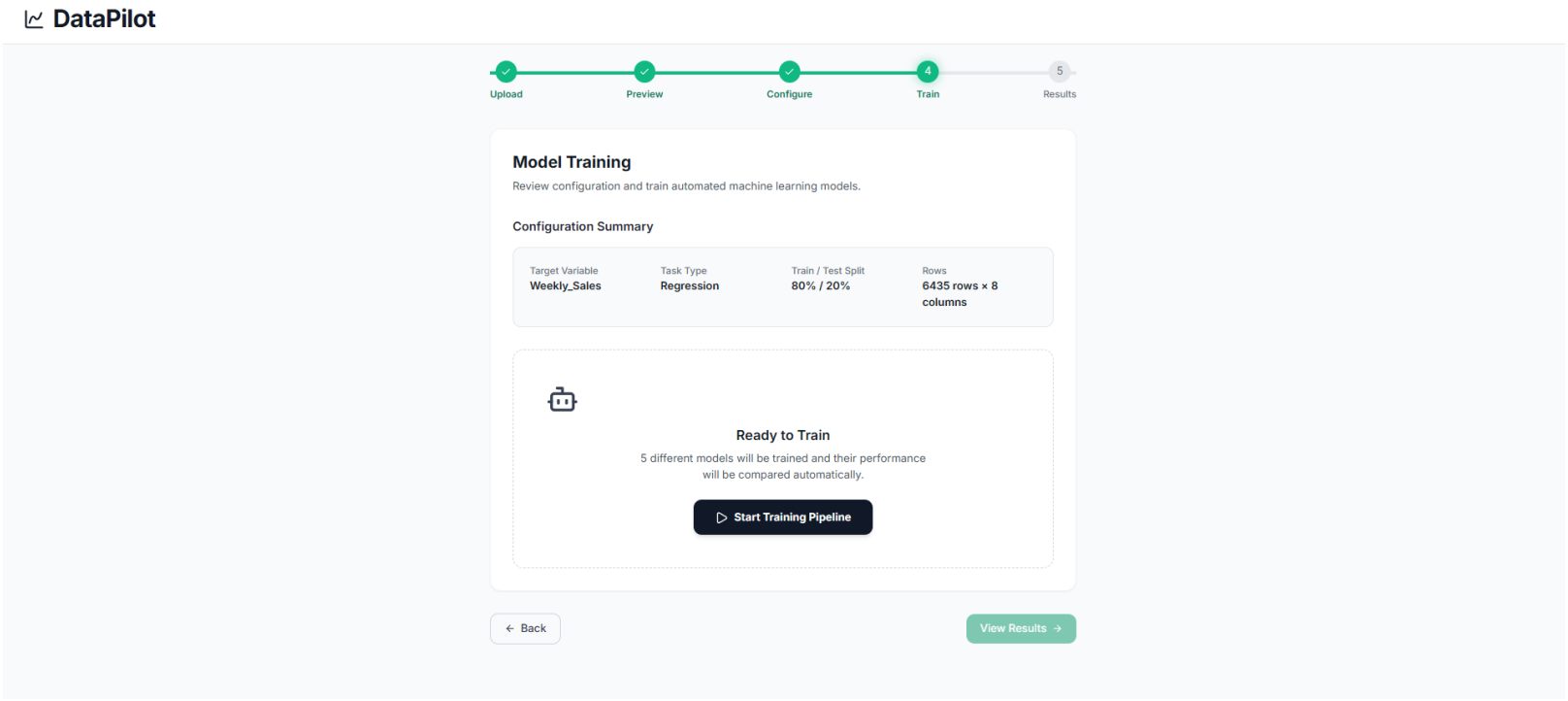


Figure 20 Model training interface

Model training is conducted automatically and after it is finished users can click on the button to view the results.

The results firstly depict the best model as well as its score based on the task type. Below that card, a bar chart is generated with each model score. For more information regarding model metrics, a leaderboard card depicts two metrics of models based on the task type. In this case since the task type is regression, the metrics used are *R2 SCORE* and *RMSE (Root Mean Squared Error)*.

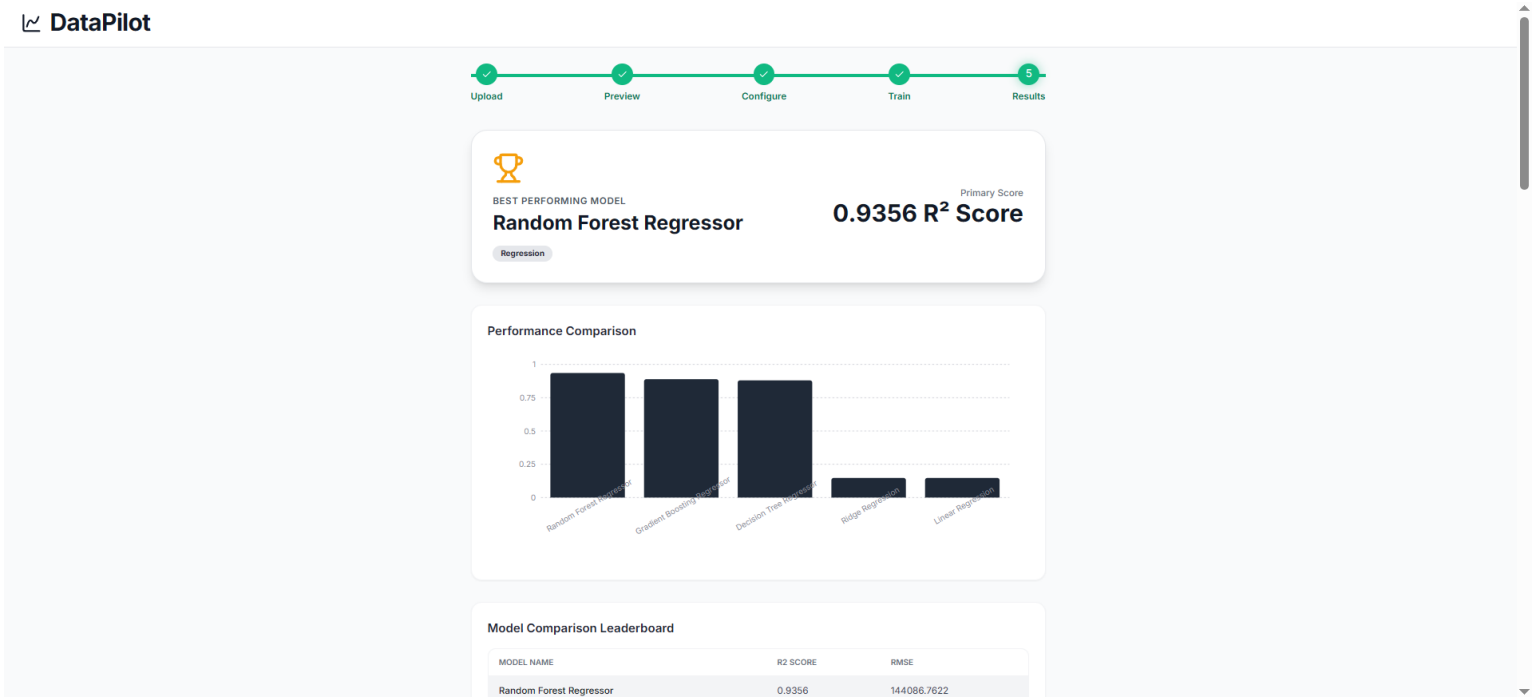


Figure 21 Performance results interface

Below the model leaderboard, the scatter plots exist for each model. These plots will be further analyzed in the testing section. In the end of this page, a correlation heatmap of the numeric features from the uploaded dataset exists for further information on how models behaved based on the data. After that, users can press the new analysis button to start over.

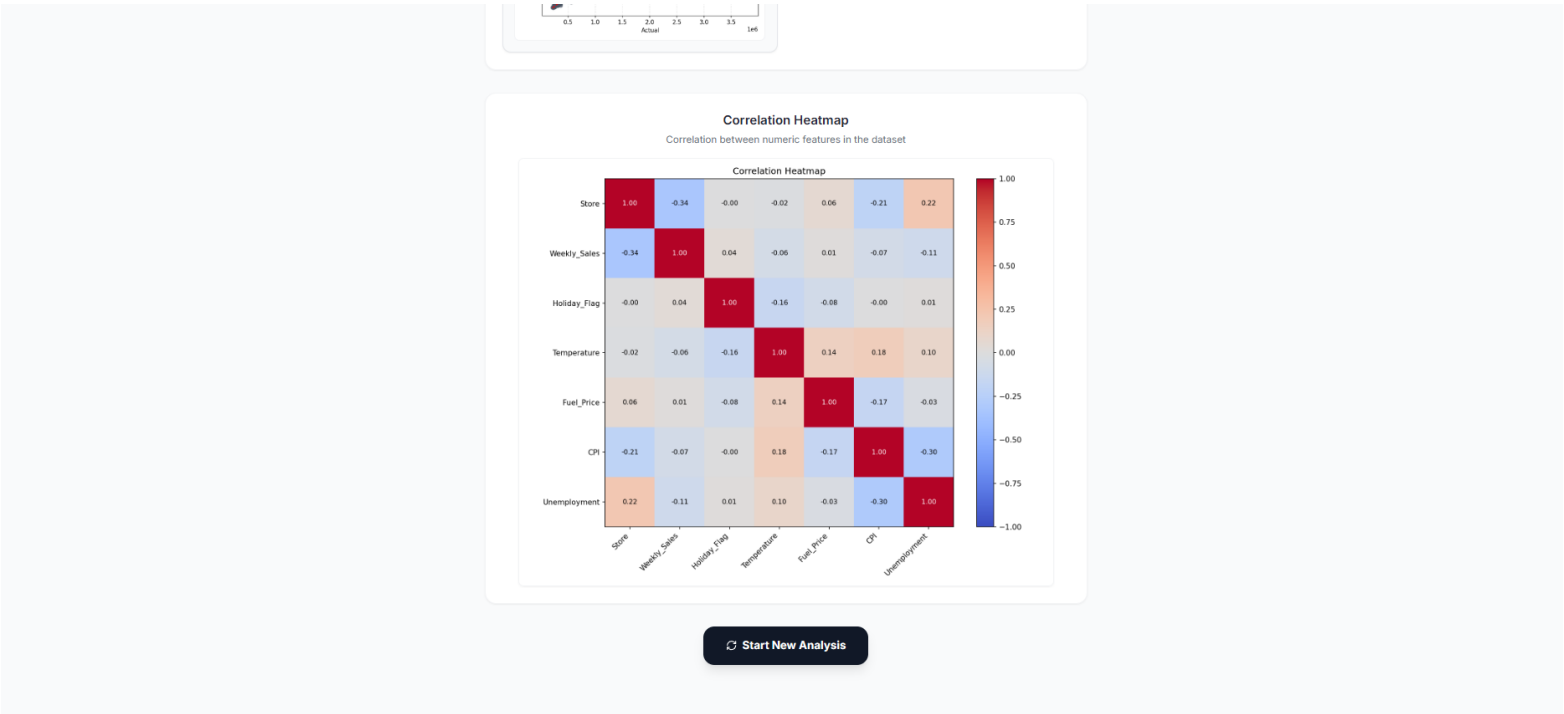


Figure 22 Performance results interface (bottom of the page)

3.5 System Testing and Evaluation

This section presents the system's tests using three different datasets. The total number of tests is four where problematic data and other alterations are tested among regular regression and classification cases.

3.5.1 Regression test

The first test conducted involves a dataset of hotel booking information (Mostipak, 2019). First the dataset is loaded, and some previews are provided by the system.

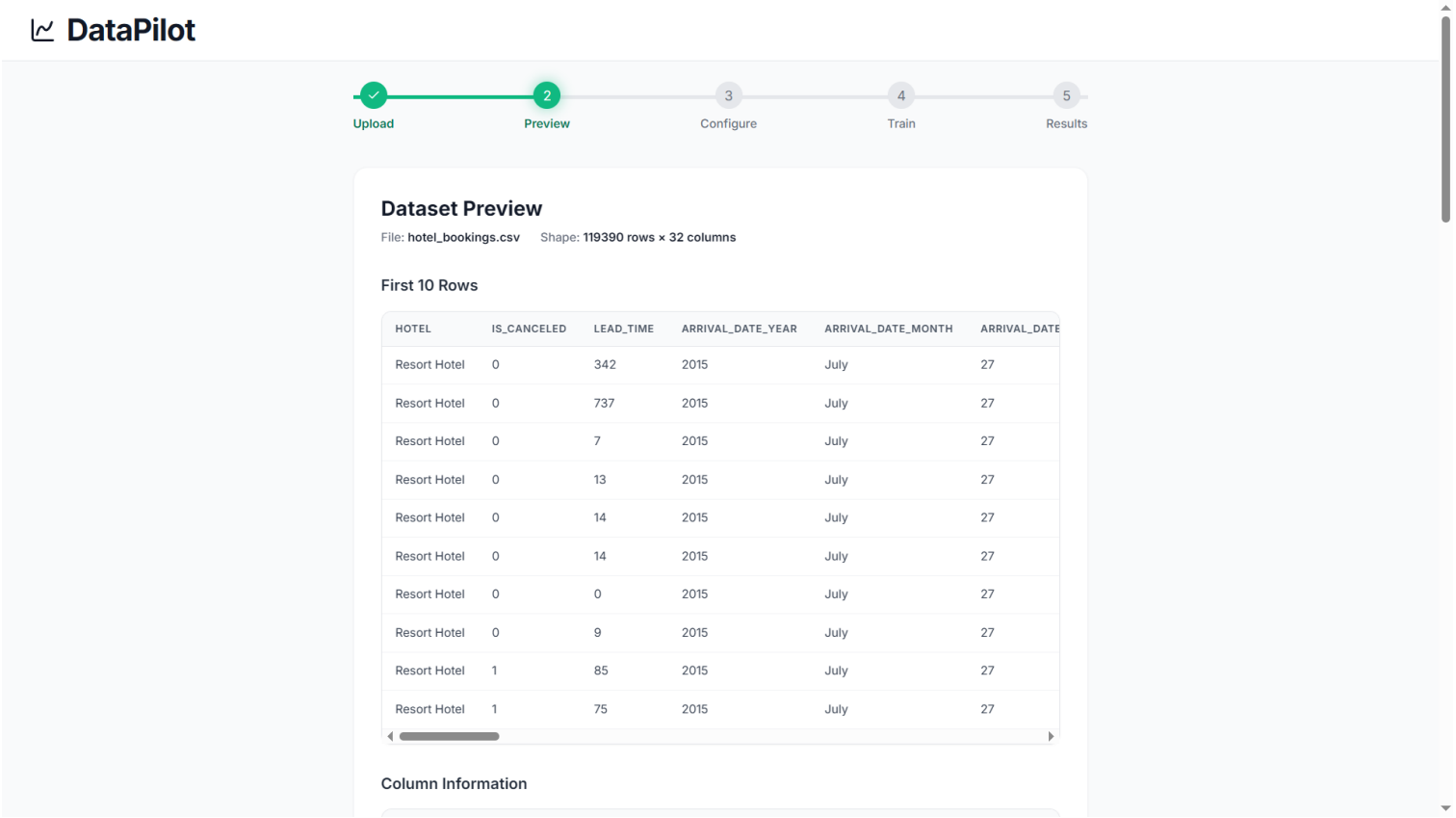


Figure 23 Regression test: Dataset preview

Column Information

COLUMN NAME	DATA TYPE	NON-NULL COUNT	UNIQUE VALUES
hotel	object	119390	2
is_canceled	int64	119390	2
lead_time	int64	119390	479
arrival_date_year	int64	119390	3
arrival_date_month	object	119390	12
arrival_date_week_number	int64	119390	53
arrival_date_day_of_month	int64	119390	31
stays_in_weekend_nights	int64	119390	17
stays_in_week_nights	int64	119390	35
adults	int64	119390	14
children	float64	119386	5
babies	int64	119390	5
meal	object	119390	5
country	object	118902	177
market_segment	object	119390	8
distribution_channel	object	119390	5
is_repeated_guest	int64	119390	2
previous_cancellations	int64	119390	15
previous_bookings_not_canceled	int64	119390	73

Figure 24 Regression test: Dataset preview (continuation)

Numeric Statistics

COLUMN	MEAN	STD DEV	MIN	MAX
is_canceled	0.3704	0.4829	0.0000	1.0000
lead_time	104.0114	106.8631	0.0000	737.0000
arrival_date_year	2016.1566	0.7075	2015.0000	2017.0000
arrival_date_week_number	27.1652	13.6051	1.0000	53.0000
arrival_date_day_of_month	15.7982	8.7808	1.0000	31.0000
stays_in_weekend_nights	0.9276	0.9986	0.0000	19.0000
stays_in_week_nights	2.5003	1.9083	0.0000	50.0000
adults	1.8564	0.5793	0.0000	55.0000
children	0.1039	0.3986	0.0000	10.0000
babies	0.0079	0.0974	0.0000	10.0000
is_repeated_guest	0.0319	0.1758	0.0000	1.0000
previous_cancellations	0.0871	0.8443	0.0000	26.0000
previous_bookings_not_canceled	0.1371	1.4974	0.0000	72.0000
booking_changes	0.2211	0.6523	0.0000	21.0000
agent	86.6934	110.7745	1.0000	535.0000
company	189.2667	131.6550	6.0000	543.0000
days_in_waiting_list	2.3211	17.5947	0.0000	391.0000
adr	101.8311	50.5358	-6.3800	5400.0000
required_car_parking_spaces	0.0625	0.2453	0.0000	8.0000

Figure 25 Regression test: Dataset preview (continuation 2)

For the regression task, the target variable is the average daily rate (adr) which involves the average income per occupied room of the hotel. By default, the system uses the train/test split of 80% to 20% so no changes will be made in that aspect.

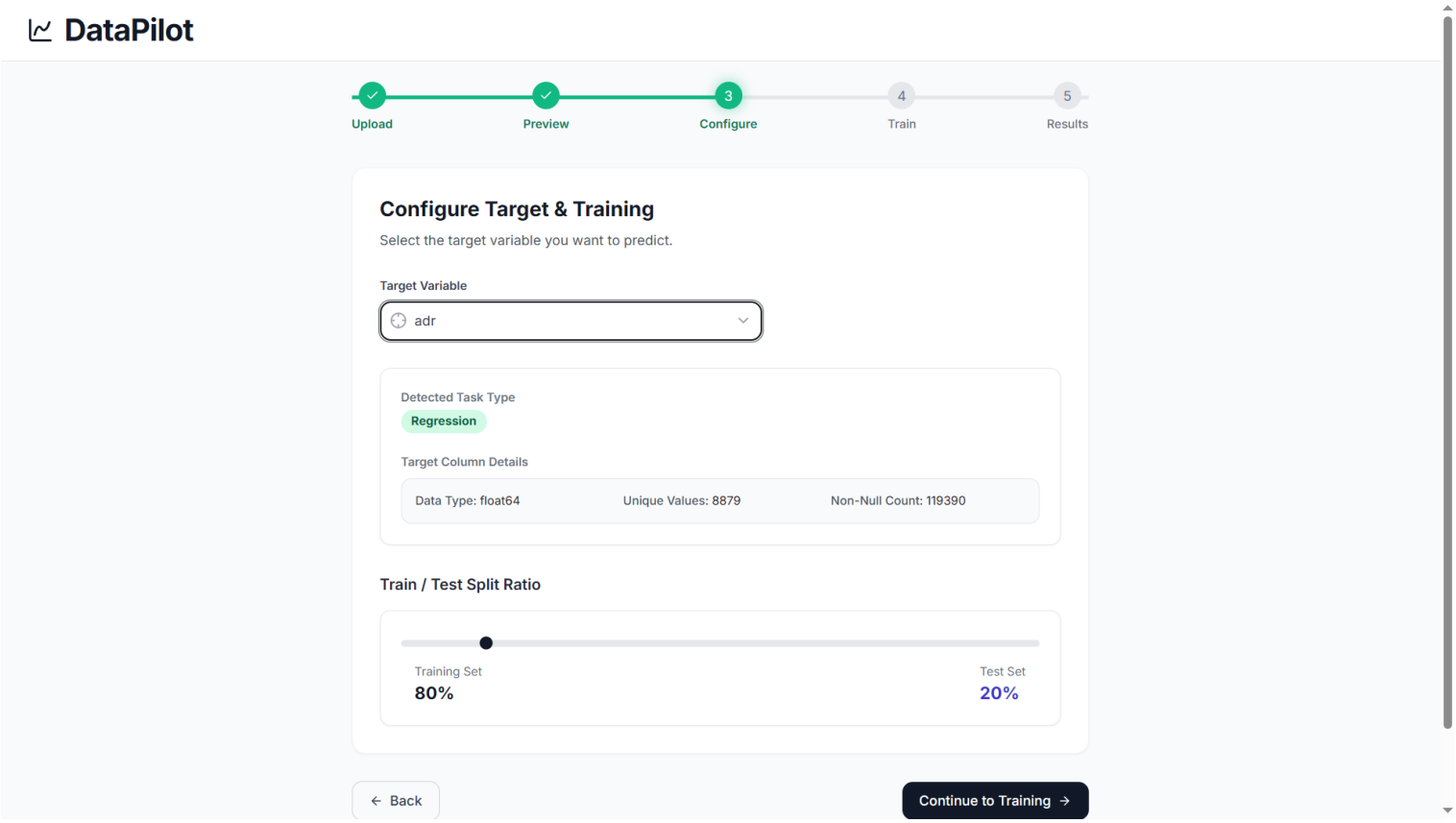


Figure 26 Regression test: Target and training configuration

After the designated models are trained the results page can be accessed. On the top of the page what can be observed is the model with top performance.

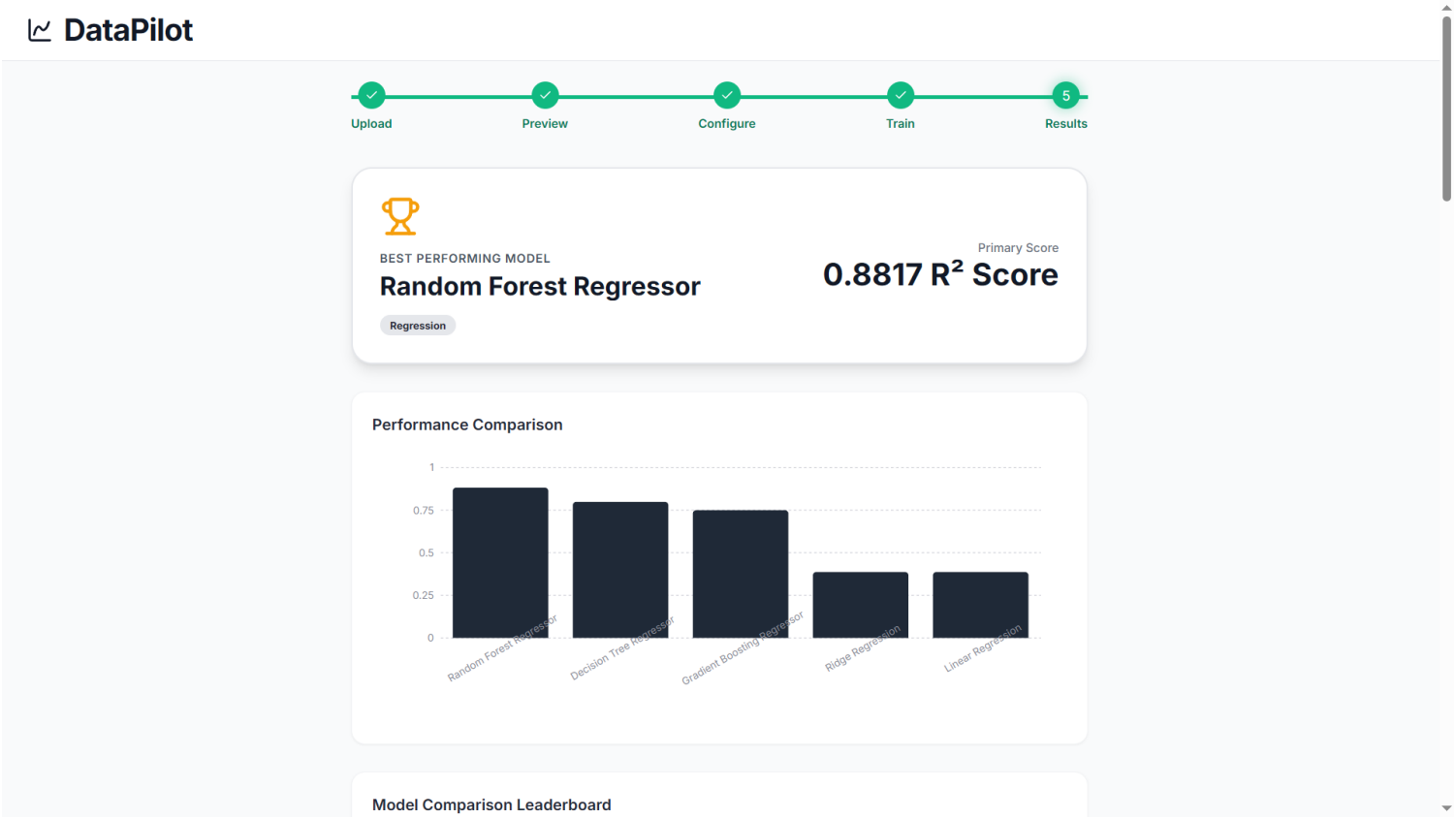


Figure 27 Regression test: Model results

In this case, the Random Forest Regressor appears to have achieved the highest score.



Figure 28 Regression test: Model results - Feature importance

With the help of the model comparison leaderboard, what can be observed is that the Random Forest model did indeed have better results than the rest of the models. Its high R2 score and low RMSE can suggest strength in the prediction of adr.

As a whole, the tree-based models appear to outperform the linear models. This implies that most relationships of the target variable are nonlinear.

Additionally, *reservation_status_date* and *arrival_date_week_number* appear to be among the top predictors of adr.

Furthermore, the following scatter plots give a better visualization of the actual performance of each model.

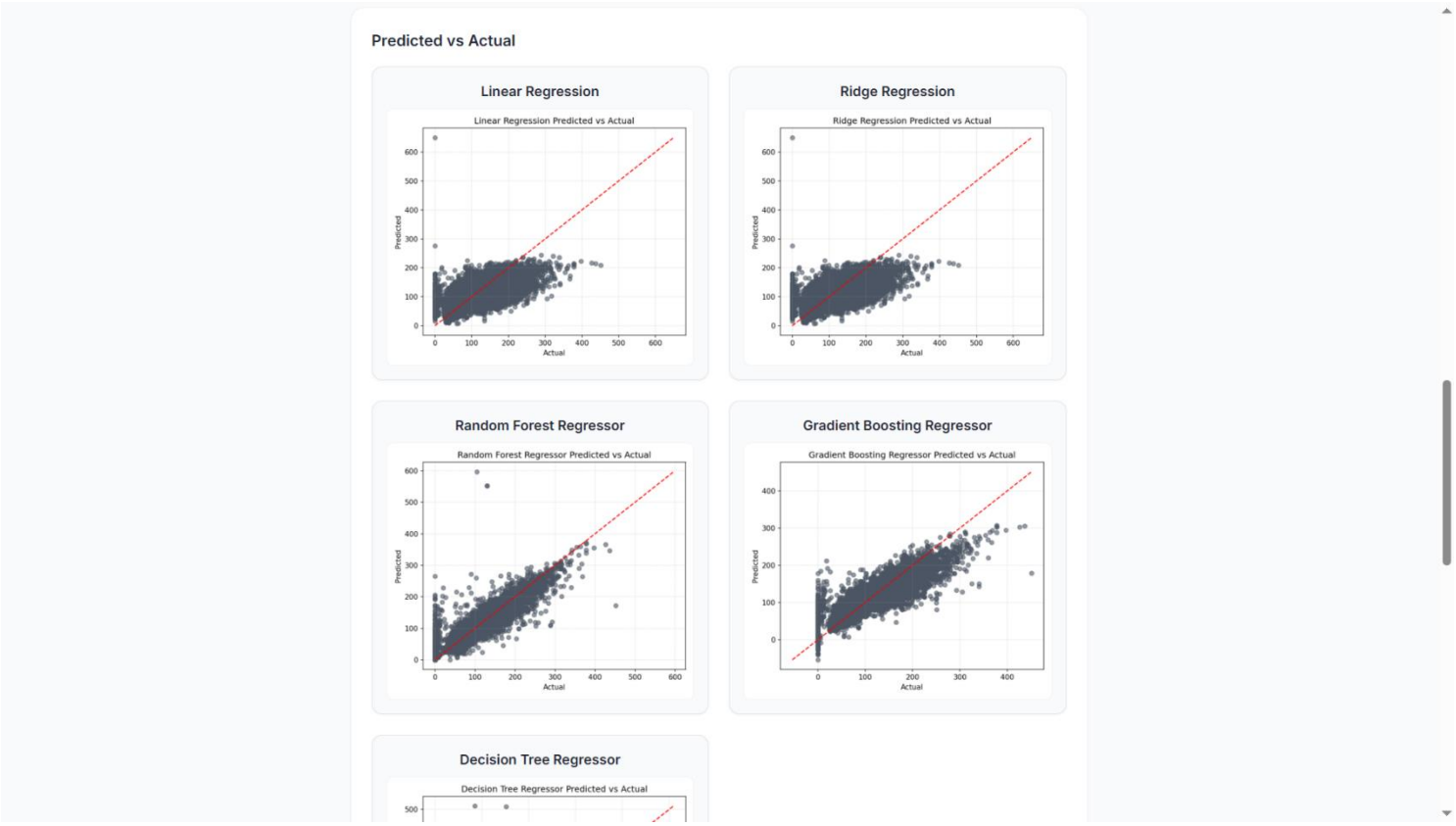


Figure 29 Regression test: Model results - Plots

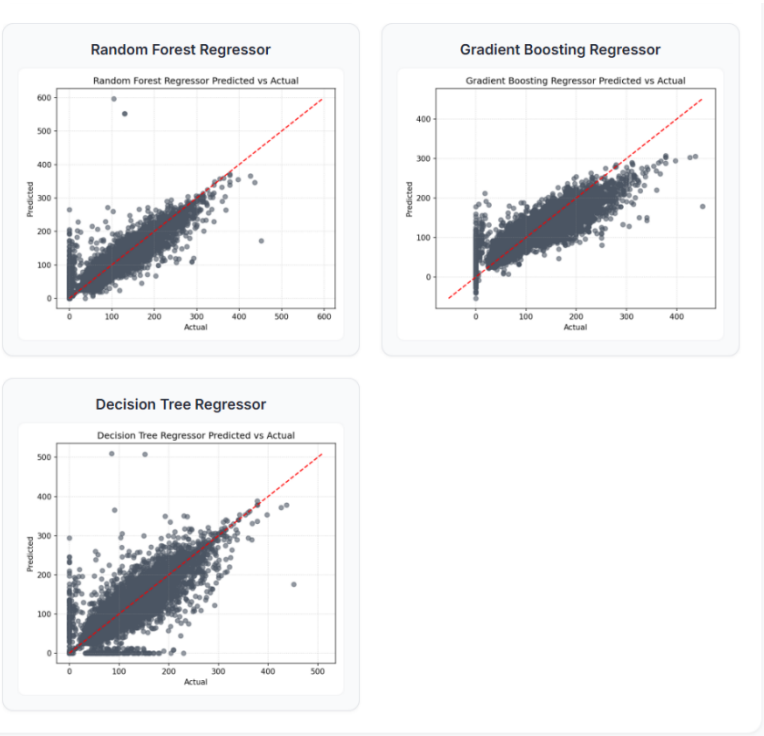


Figure 30 Regression test: Model results - Plots (continuation)

The plots support the previous results. The top performed models appear to have predicted closer to the actual values. This can be observed on the plots of Random Forest Regressor and Gradient Boosting Regressor.

3.5.2 Classification Test

For the classification test, a dataset of Titanic survival prediction is used (M_Yasser_H, 2022).

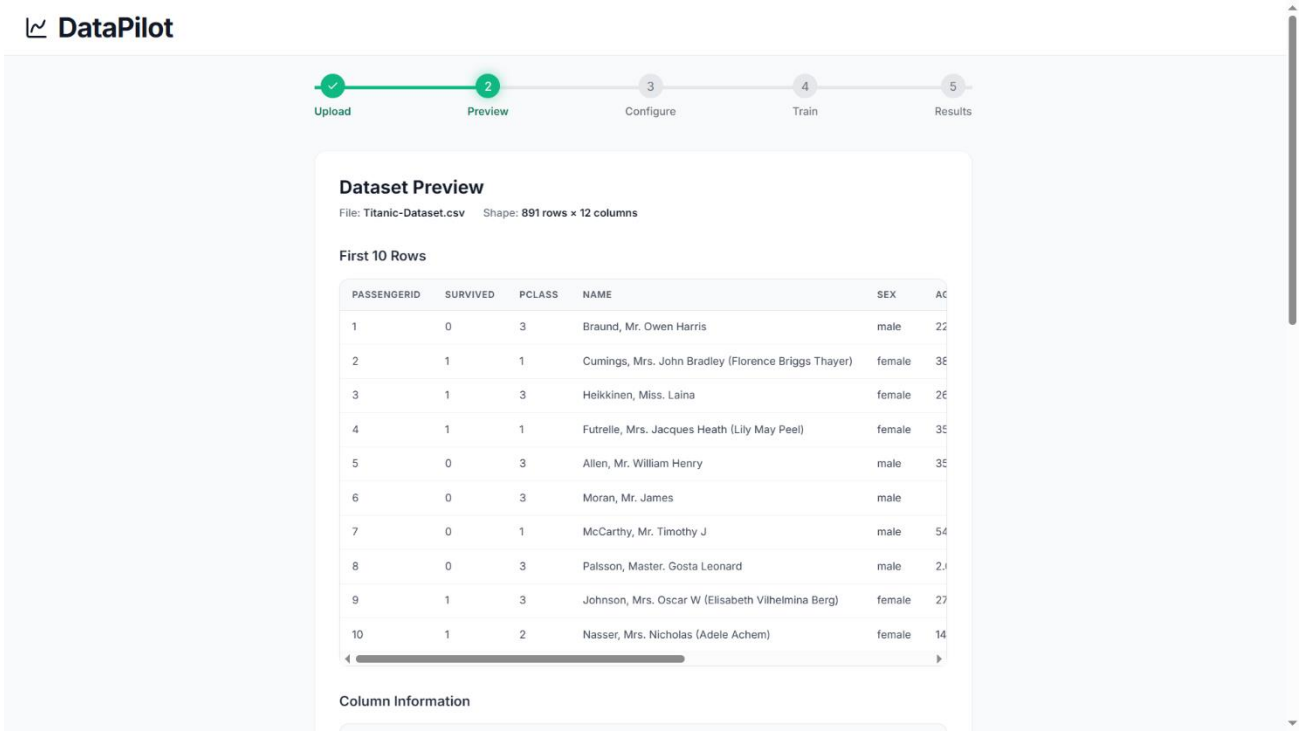
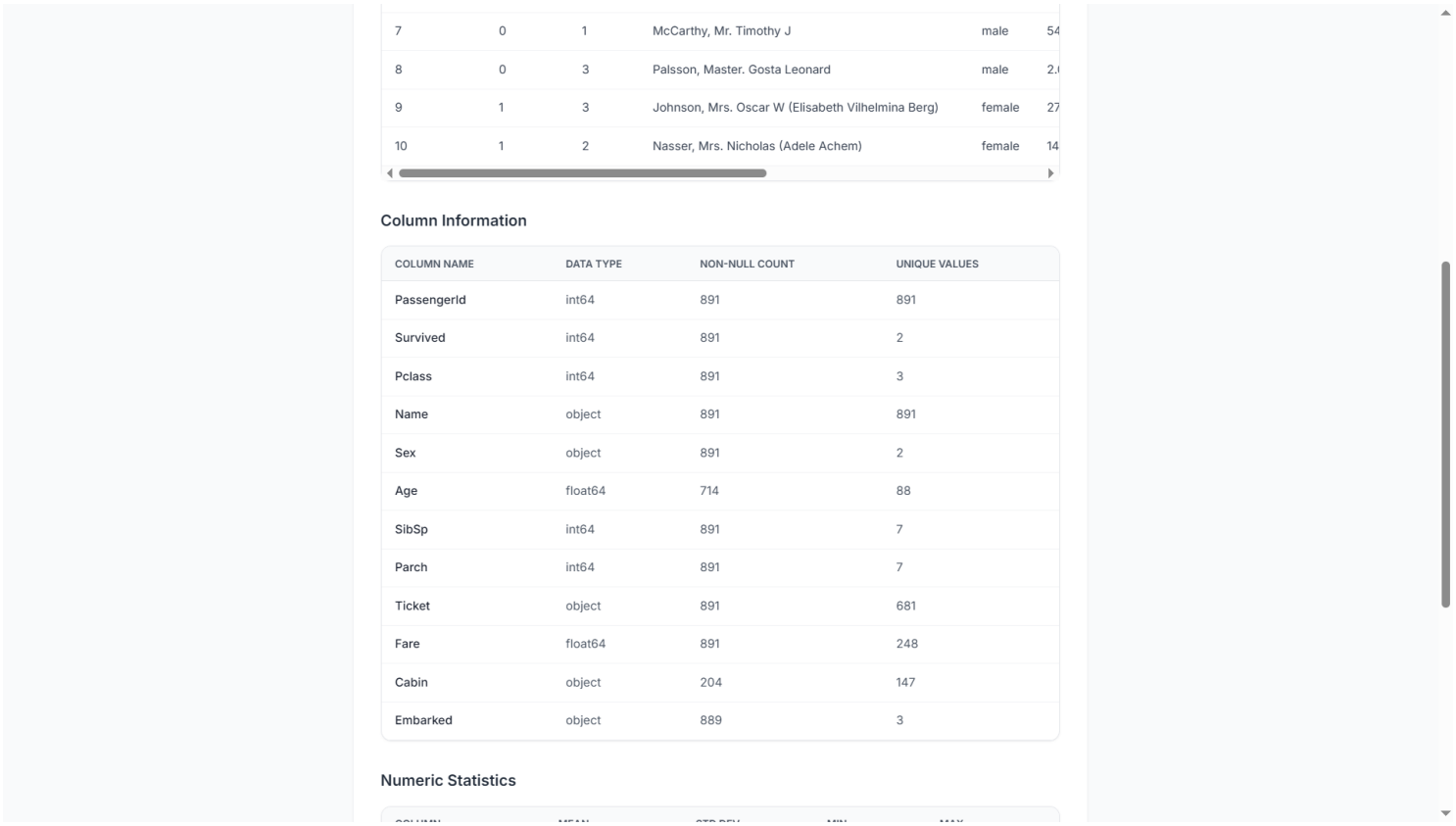


Figure 31 Classification Test: Dataset preview



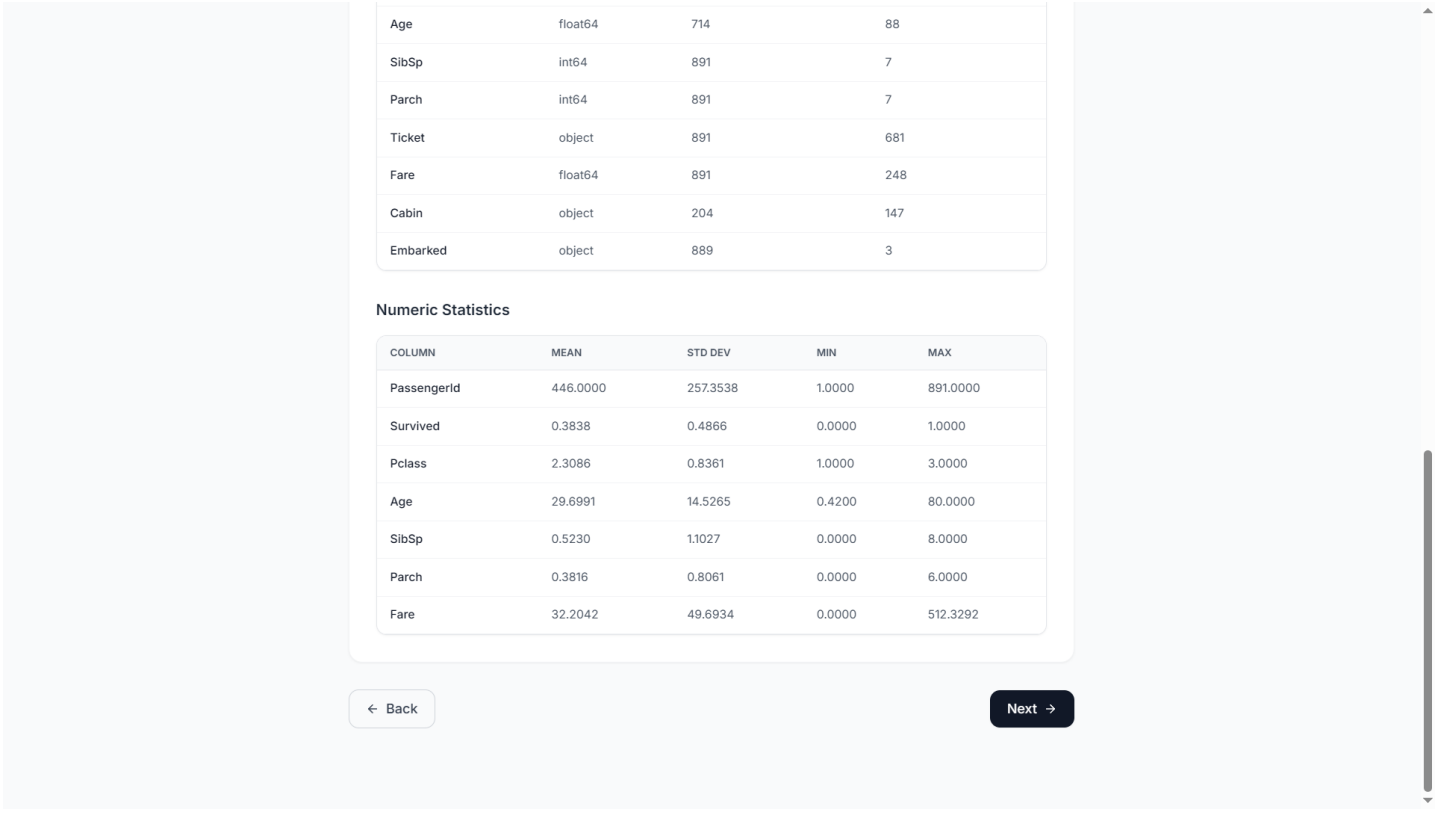
7	0	1	McCarthy, Mr. Timothy J	male	54
8	0	3	Palsson, Master. Gosta Leonard	male	2.
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14

Column Information

COLUMN NAME	DATA TYPE	NON-NULL COUNT	UNIQUE VALUES
PassengerId	int64	891	891
Survived	int64	891	2
Pclass	int64	891	3
Name	object	891	891
Sex	object	891	2
Age	float64	714	88
SibSp	int64	891	7
Parch	int64	891	7
Ticket	object	891	681
Fare	float64	891	248
Cabin	object	204	147
Embarked	object	889	3

Numeric Statistics

Figure 32 Classification Test: Dataset preview (continuation)



Age	float64	714	88
SibSp	int64	891	7
Parch	int64	891	7
Ticket	object	891	681
Fare	float64	891	248
Cabin	object	204	147
Embarked	object	889	3

Numeric Statistics

COLUMN	MEAN	STD DEV	MIN	MAX
PassengerId	446.0000	257.3538	1.0000	891.0000
Survived	0.3838	0.4866	0.0000	1.0000
Pclass	2.3086	0.8361	1.0000	3.0000
Age	29.6991	14.5265	0.4200	80.0000
SibSp	0.5230	1.1027	0.0000	8.0000
Parch	0.3816	0.8061	0.0000	6.0000
Fare	32.2042	49.6934	0.0000	512.3292

← Back

Next →

Figure 33 Classification Test: Dataset preview (continuation 2)

The dataset contains information about passenger details. For the case of classification, the *Survived* variable can be used for the prediction of survival in the Titanic.

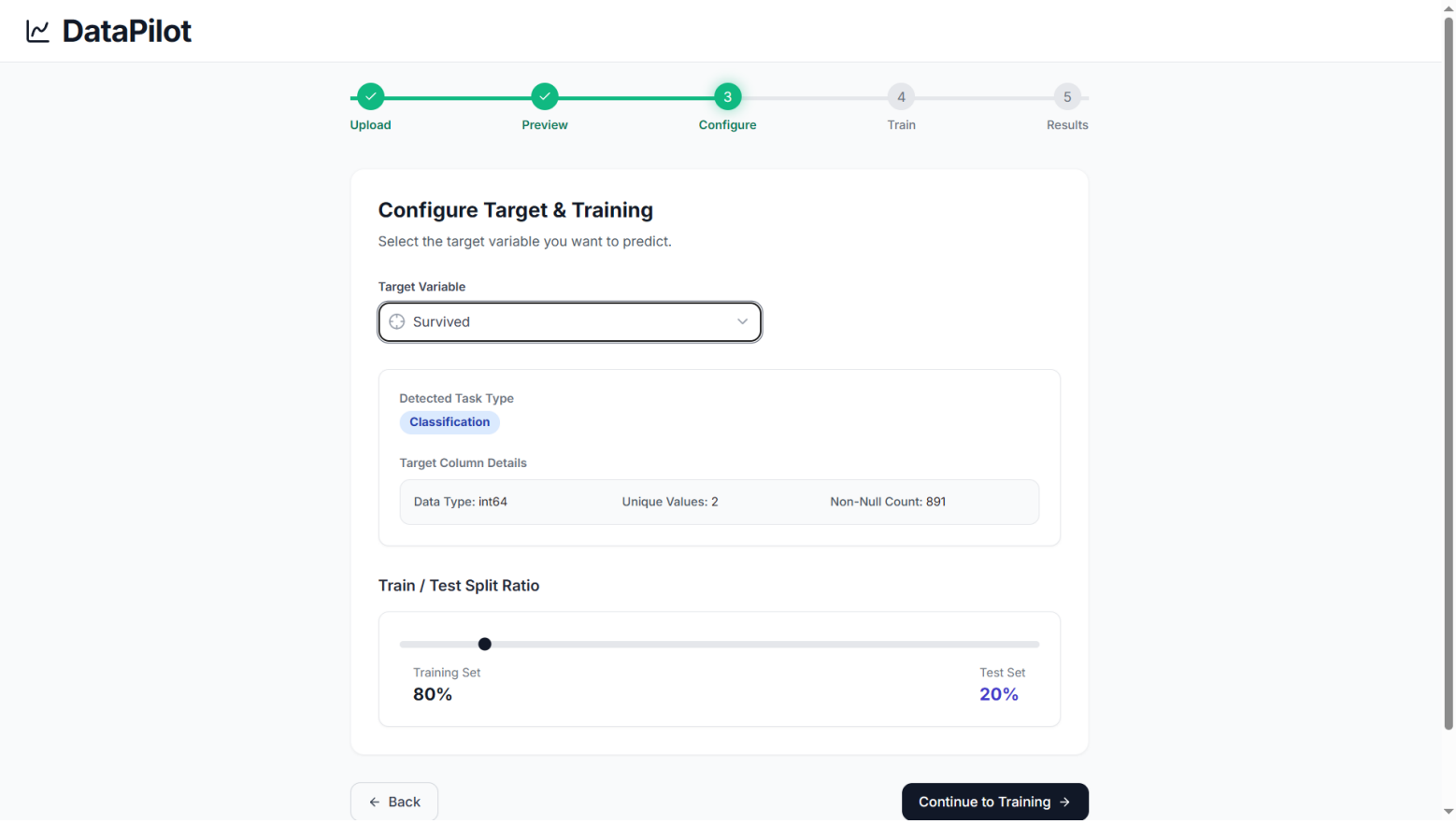


Figure 34 Classification Test: Target and training configuration

The target variable appears to be a binary conclusion of whether a passenger has survived or not. As previously applied, the train/test split remains at its default values by the system.

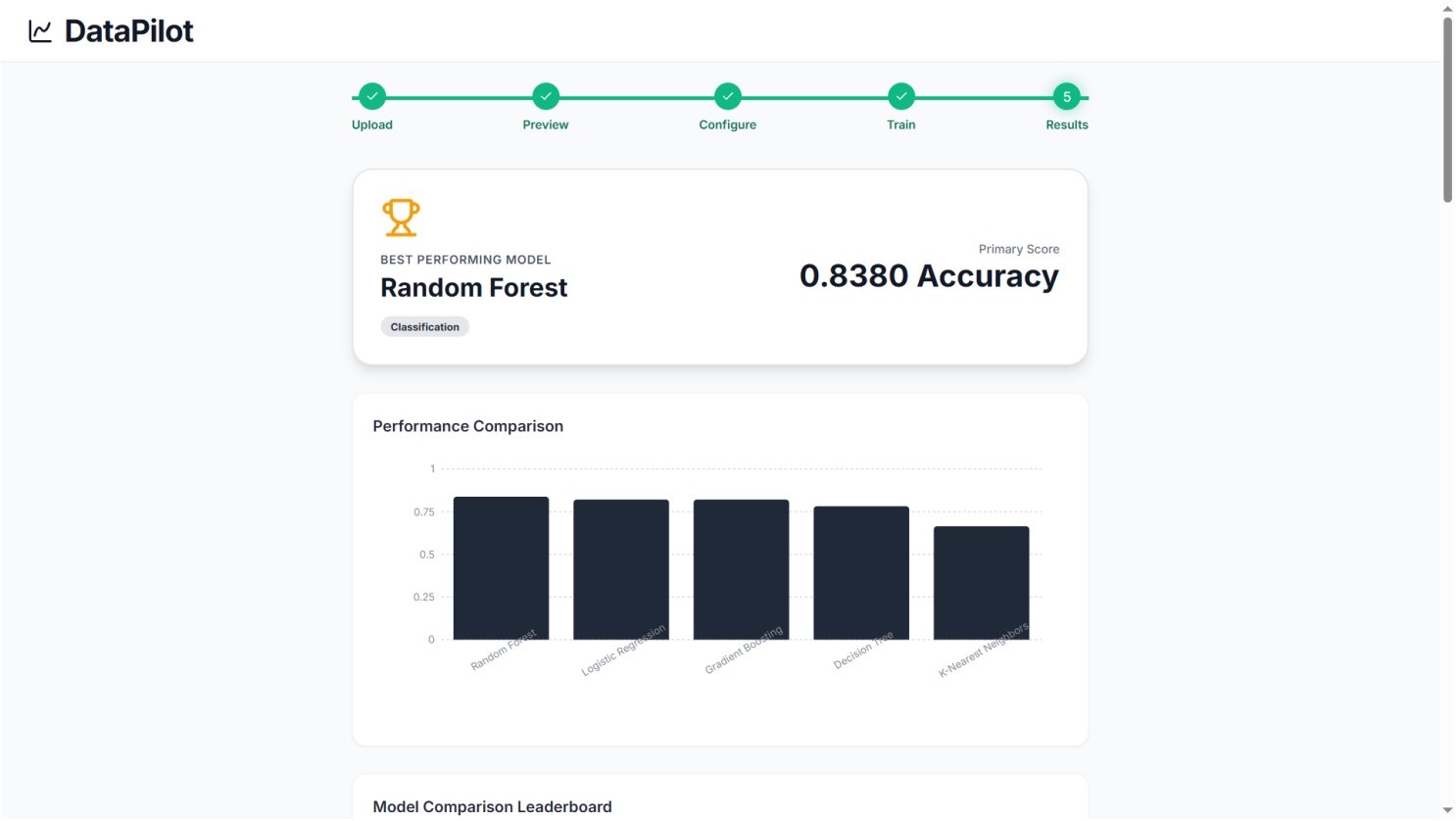


Figure 35 Classification Test: Model results

The best model appears to be Random Forest which achieved an accuracy of 0.83. From the performance comparison bar chart, it appears that all five models performed relatively well and seem to be close in terms of scoring.



Figure 36 Classification Test: Model results - Feature importance

The Random Forest model achieved the highest scores compared to the other models. Despite that, the scores of Logistic Regression and Gradient Boosting appear to be very close to those of the winning model. Concurrently, the K-Nearest Neighbors model achieved very low scores. This can be caused because the dataset used contains mixed or uneven features that the KNN model might not be able to learn well from. The feature importance graph suggests that the sex of passengers appears to be the strongest predictor of survival.

The following figures of model confusion matrices can help in a clearer identification of the classification model behavior.

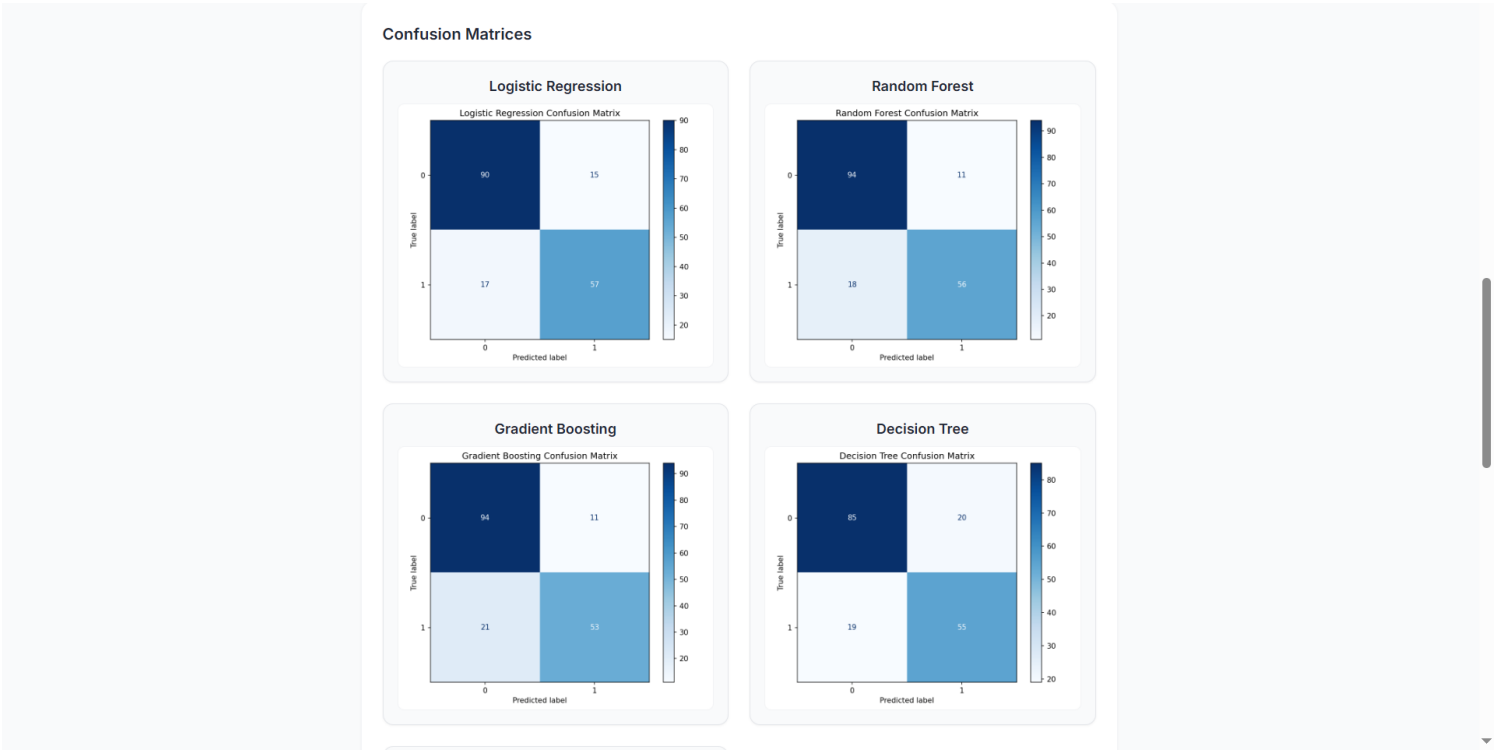


Figure 37 Classification Test: Model results - Confusion matrices



Figure 38 Classification Test: Model results - Confusion matrices (continuation)

The confusion matrices validate the findings of the analysis. The Random Forest, Logistic Regression and Gradient Boosting models appear to have made the most correct predictions. The Decision Tree model achieved a similarly good performance while the K-Nearest Neighbor has more incorrect predictions, especially in false negatives and false positives which can support the previous assumption as to the model's behavior.

3.5.3 Train/Test Split Testing

The following test will evaluate how the interactive Train/Test Split bar will affect an analysis if alternations are made. For this test, a dataset regarding Walmart sales is used (Mikhail, 2024).

DataPilot

Upload 2 Preview 3 Configure 4 Train 5 Results

Dataset Preview

File: **Walmart_Sales.csv** Shape: **6435 rows × 8 columns**

First 10 Rows

STORE	DATE	WEEKLY_SALES	HOLIDAY_FLAG	TEMPERATURE	FUEL_PRICE	CPI
1	05-02-2010	1643690.9	0	42.31	2.572	211.0963582
1	12-02-2010	1641957.44	1	38.51	2.548	211.2421698
1	19-02-2010	1611968.17	0	39.93	2.514	211.2891429
1	26-02-2010	1409727.59	0	46.63	2.561	211.3196429
1	05-03-2010	1554806.68	0	46.5	2.625	211.3501429
1	12-03-2010	1439541.59	0	57.79	2.667	211.3806429
1	19-03-2010	1472515.79	0	54.58	2.72	211.215635
1	26-03-2010	1404429.92	0	51.45	2.732	211.0180424
1	02-04-2010	1594968.28	0	62.27	2.719	210.8204499
1	09-04-2010	1545418.53	0	65.86	2.77	210.6228574

Column Information

Figure 39 Train/Test Split Testing: Data preview

Column Information

COLUMN NAME	DATA TYPE	NON-NULL COUNT	UNIQUE VALUES
Store	int64	6435	45
Date	object	6435	143
Weekly_Sales	float64	6435	6435
Holiday_Flag	int64	6435	2
Temperature	float64	6435	3528
Fuel_Price	float64	6435	892
CPI	float64	6435	2145
Unemployment	float64	6435	349

Numeric Statistics

COLUMN	MEAN	STD DEV	MIN	MAX
Store	23.0000	12.9882	1.0000	45.0000
Weekly_Sales	1046964.8776	564366.6221	209986.2500	3818686.4500
Holiday_Flag	0.0699	0.2550	0.0000	1.0000
Temperature	60.6638	18.4449	-2.0600	100.1400
Fuel_Price	3.3586	0.4590	2.4720	4.4680
CPI	171.5784	39.3567	126.0640	227.2328
Unemployment	7.9992	1.8759	3.8790	14.3130

Figure 40 Train/Test Split Testing: Data preview (continuation)

In order to view any effects of this test, the same target variable has been analyzed in the default and alternative split conditions.

A comparison of the two test conditions is followed with the alternative split conditions of 65% to 35% being depicted first.

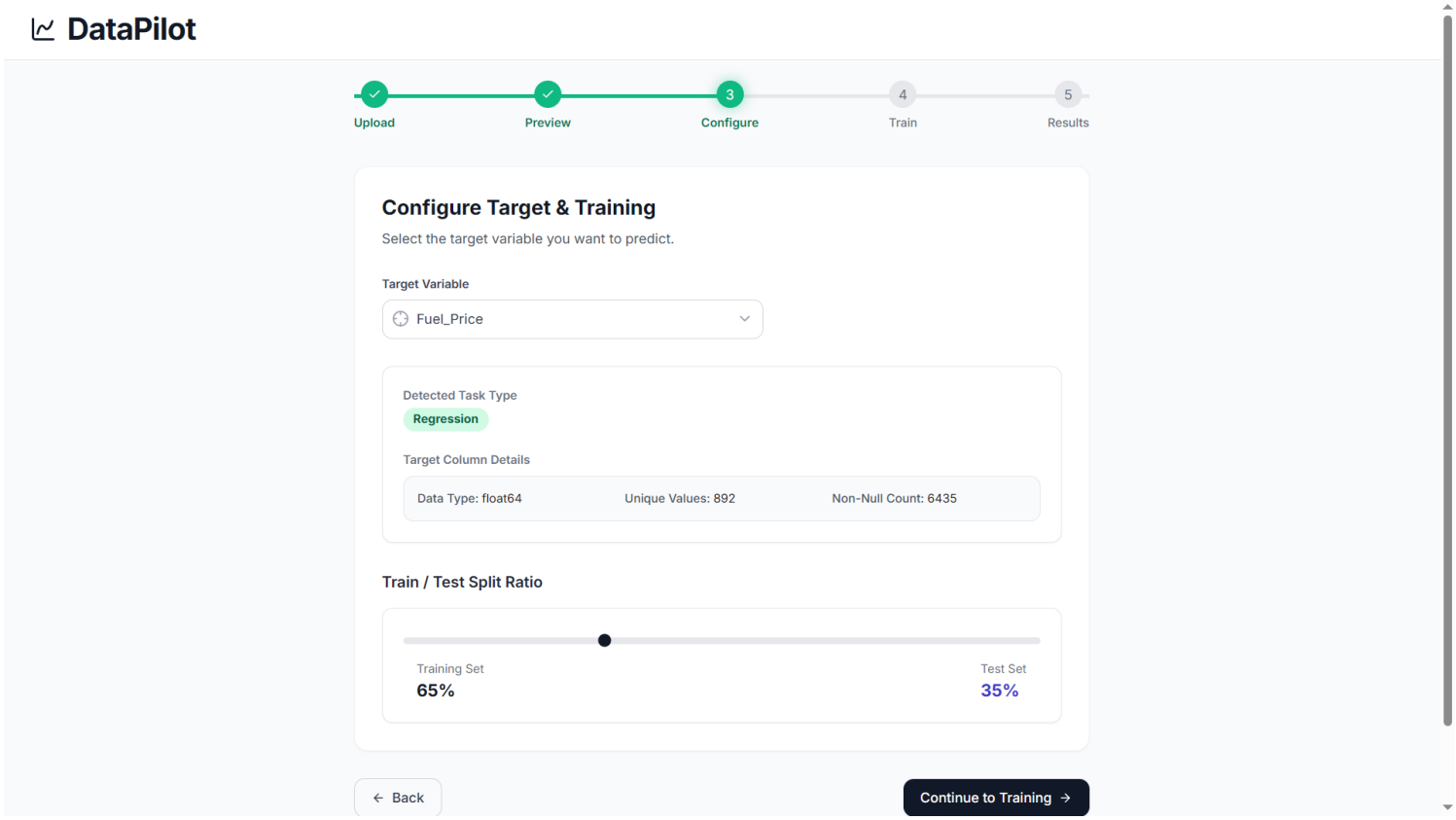


Figure 41 Train/Test Split Testing: Training configuration using the experimental setup

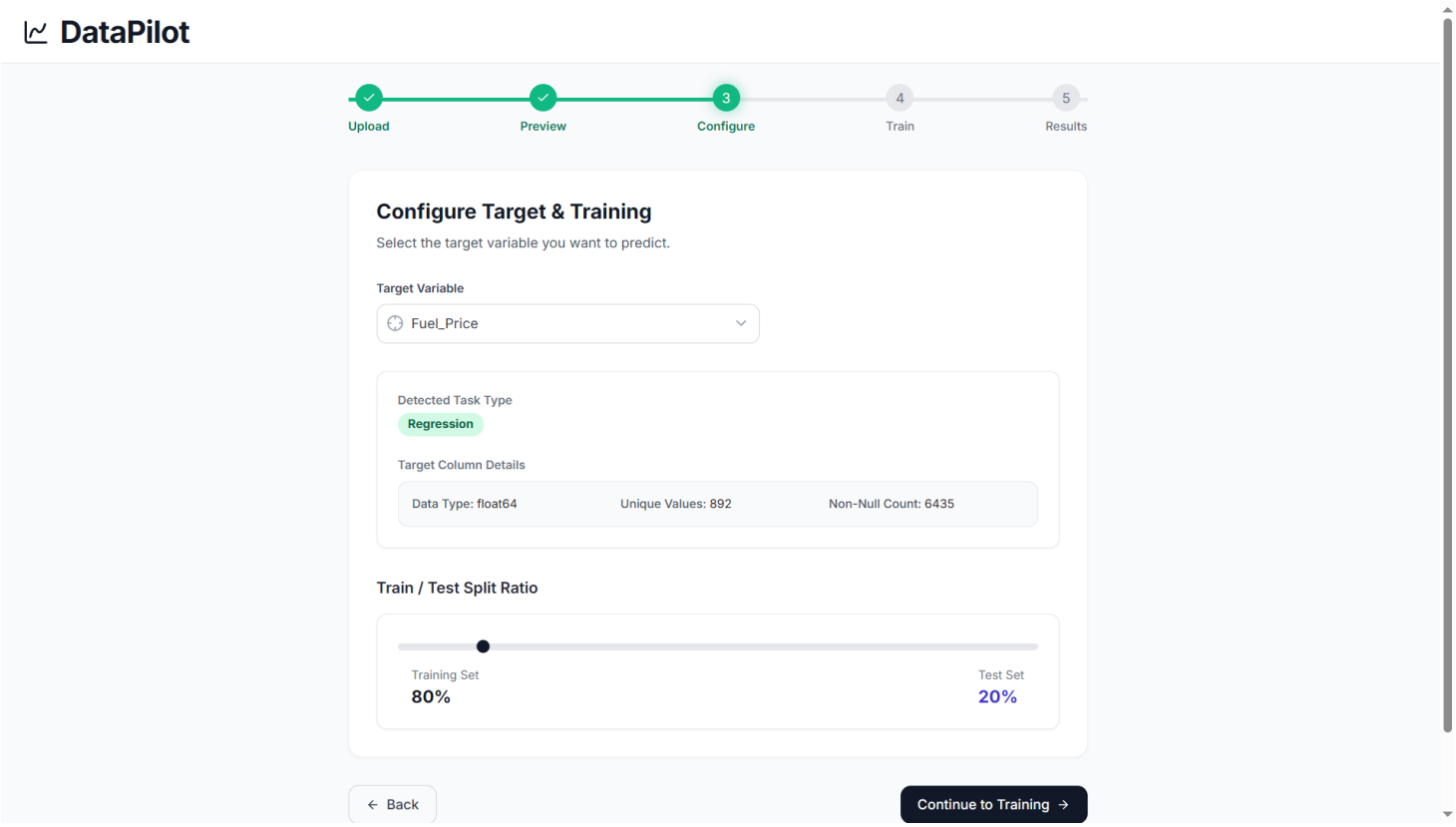


Figure 42 Train/Test Split Testing: Training configuration using the default setup

Like in all cases, during the model training phase, a configuration summary exists that gives information of some previously set aspects of the analysis.

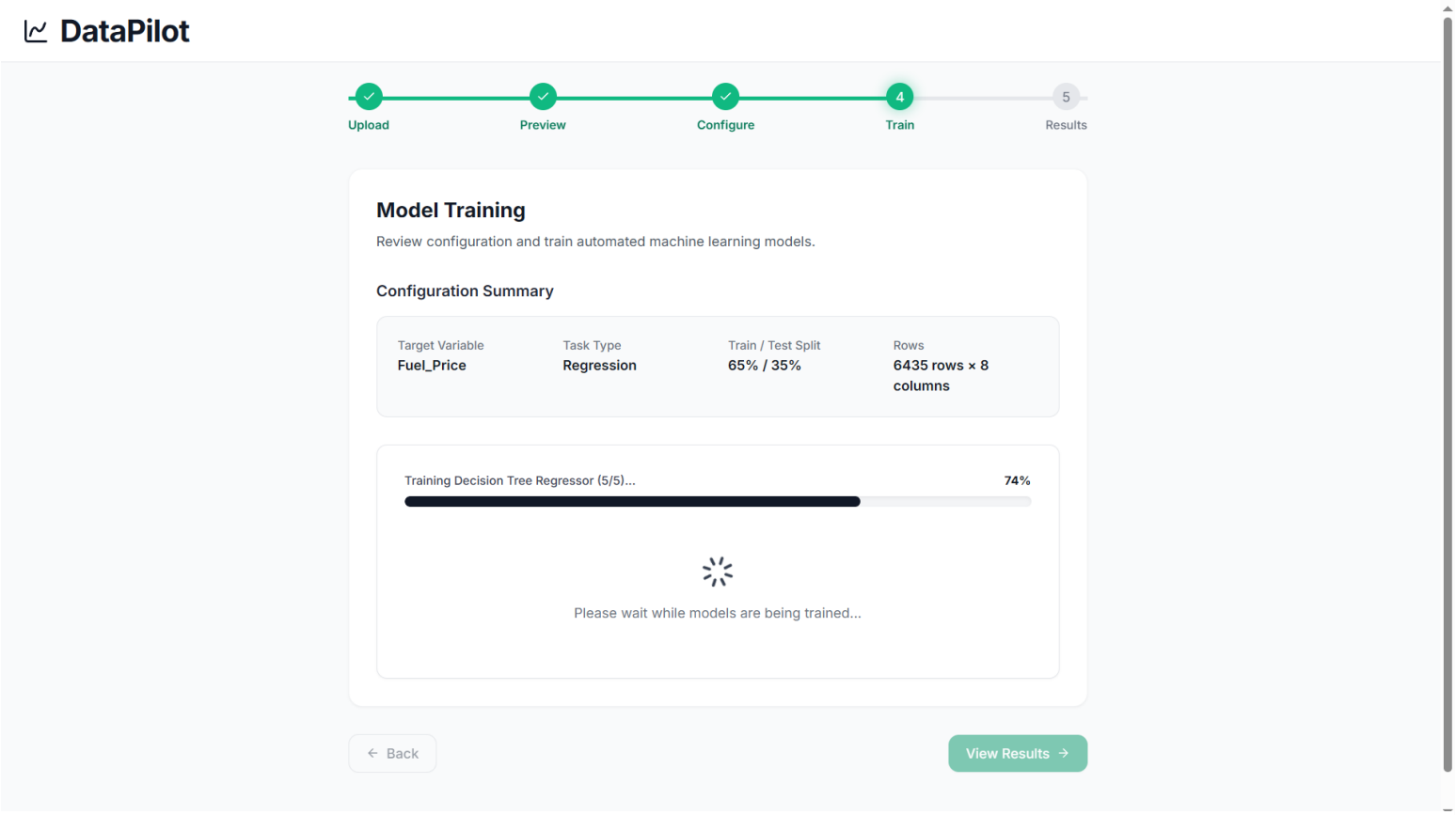


Figure 43 Train/Test Split Testing: Configuration summary

The configuration card correctly depicts the set split for the analysis. Below the results of the alternative split are depicted followed by the results of the default set.

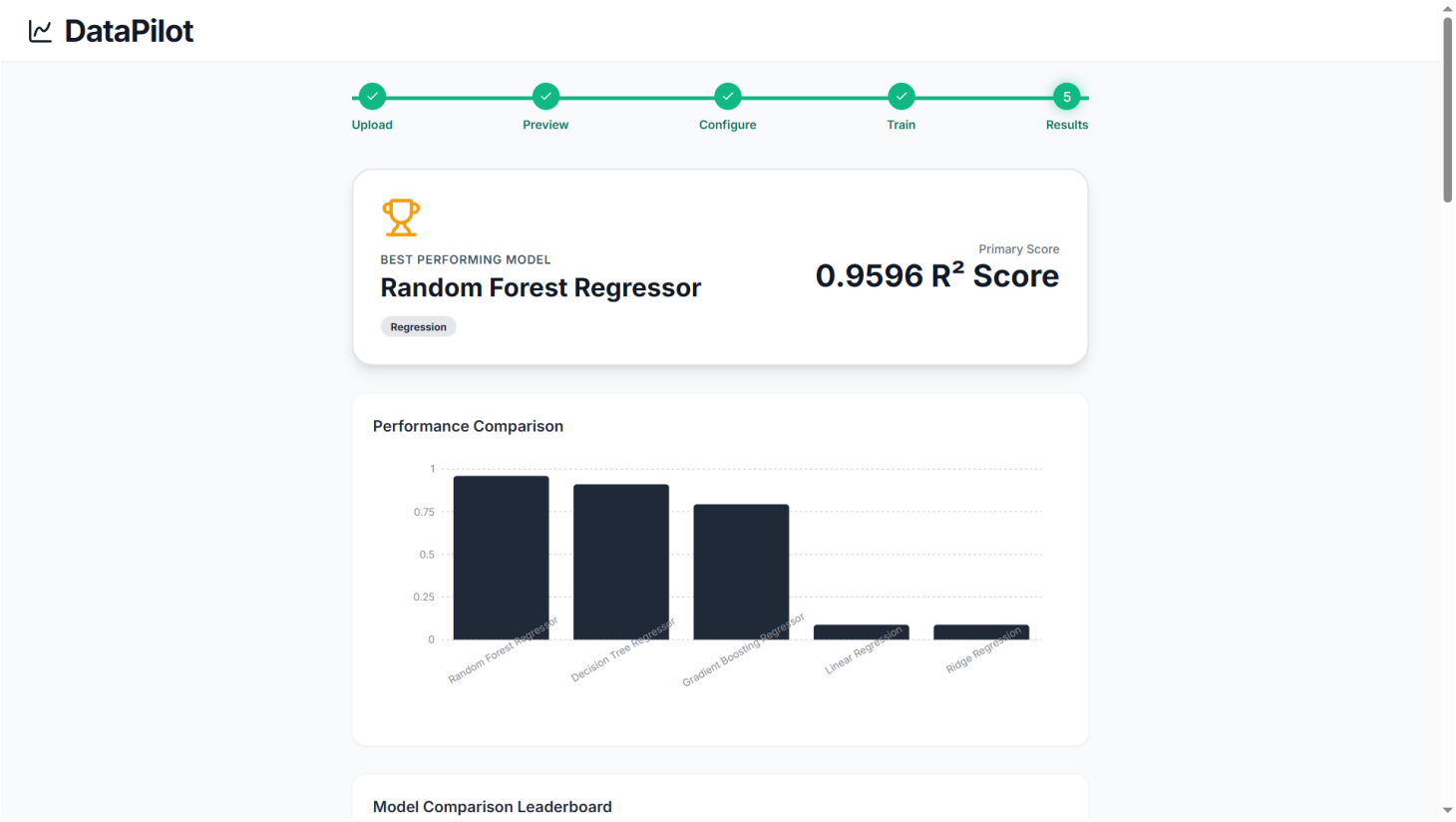


Figure 44 Train/Test Split Testing: Model results



Figure 45 Train/Test Split Testing: Model results - Feature importance

The Random Forest Regressor model appears to be the best fit in this analysis. The Decision Tree and Gradient Boosting models show a good performance whereas Linear and Ridge Regression achieved low scores. This suggests that some nonlinearity exists. CPI appears to be the most influential predictor in the case of fuel pricing. Before the default set results are presented, the depiction of the scatter plots can provide further information.



Figure 46 Train/Test Split Testing: Model results - Plots

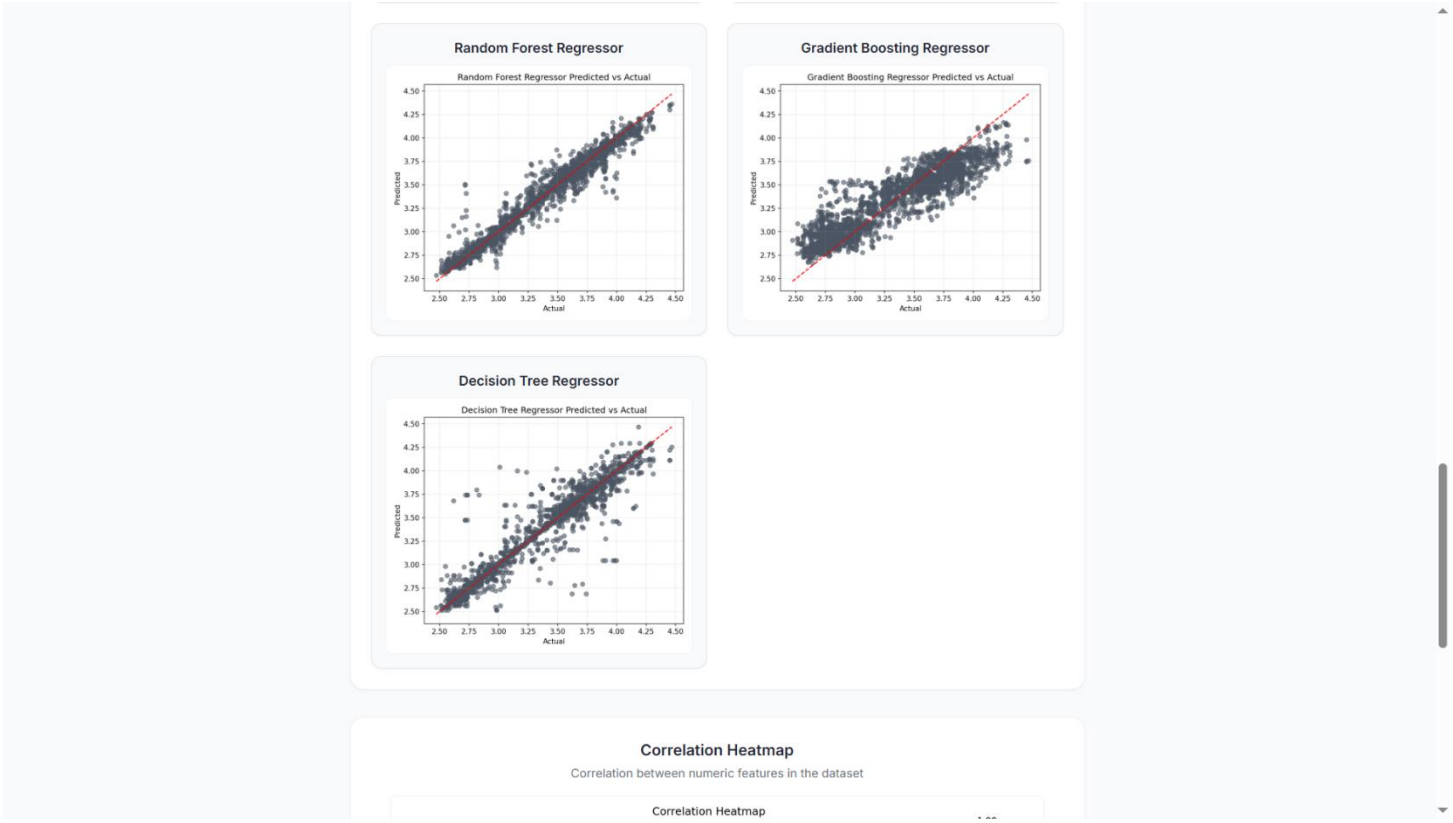


Figure 47 Train/Test Split Testing: Model results - Plots (continuation)

The plots support the earlier scores and provide a better understanding of model behavior. The top three models appear to have made predictions closer to reality.

The following results are from the default 80% to 20% split and are provided for further comparison.



Figure 48 Train/Test Split Testing: Results of the default split analysis

With the 80/20 split the models achieved higher performance since more training data was available. The Random Forest Regressor remained the best performing model with Decision Tree and Gradient Boosting models coming next. The change in results is not large but by comparing the two split results, it appears that the system is able to correctly perform differentiations in train and testing split which advocates for additional reliability of the model evaluation process.

3.5.4 Data Validity Test

The system is provided with an integrated check that exists for the user's acknowledgement of possible data leakage, overfitting or invalid parts of data from the uploaded dataset. In this test the dataset regarding hotel booking information is reused.

As depicted below, the dataset contains the *reservation_status* variable that can leak information based on the cancellation of a booking.

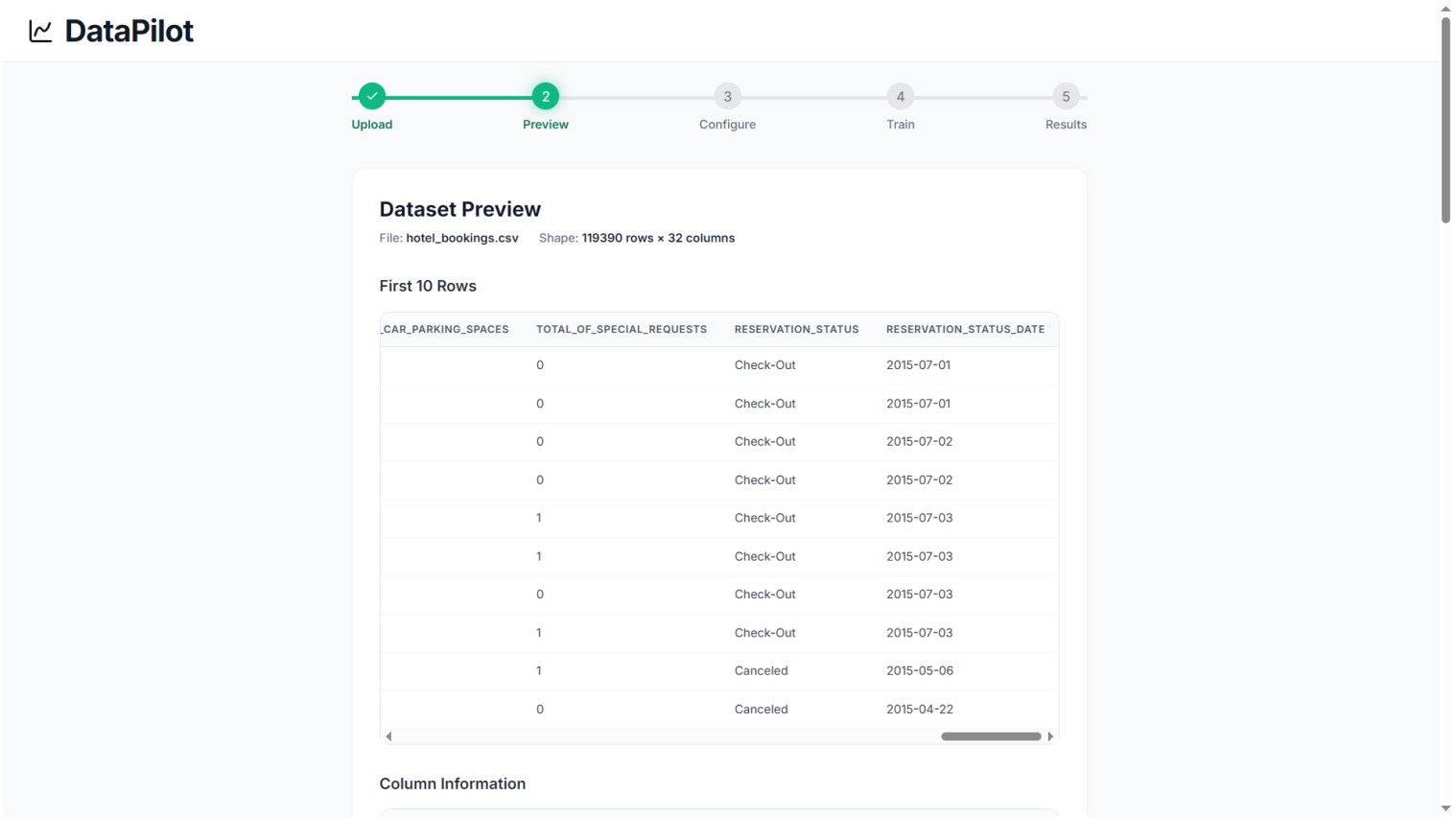


Figure 49 Data Validity Test: Data preview

For this reason, the target variable of this analysis will be the classification of whether a booking has been cancelled or not.

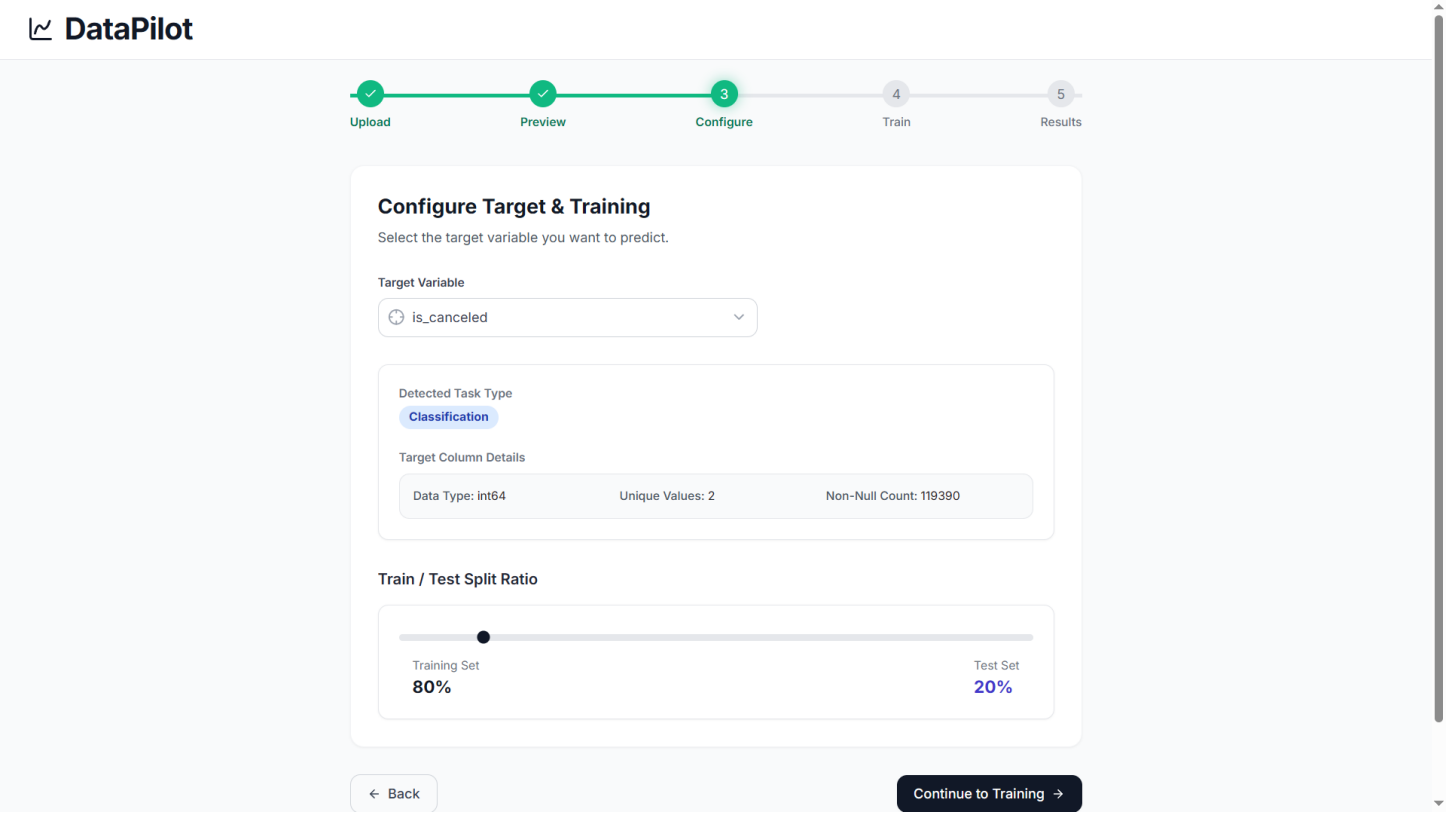


Figure 50 Data Validity Test: Training configuration

After model training, the results appear as in all cases. This time however, a message appears at the top of the page that warns about data validity.

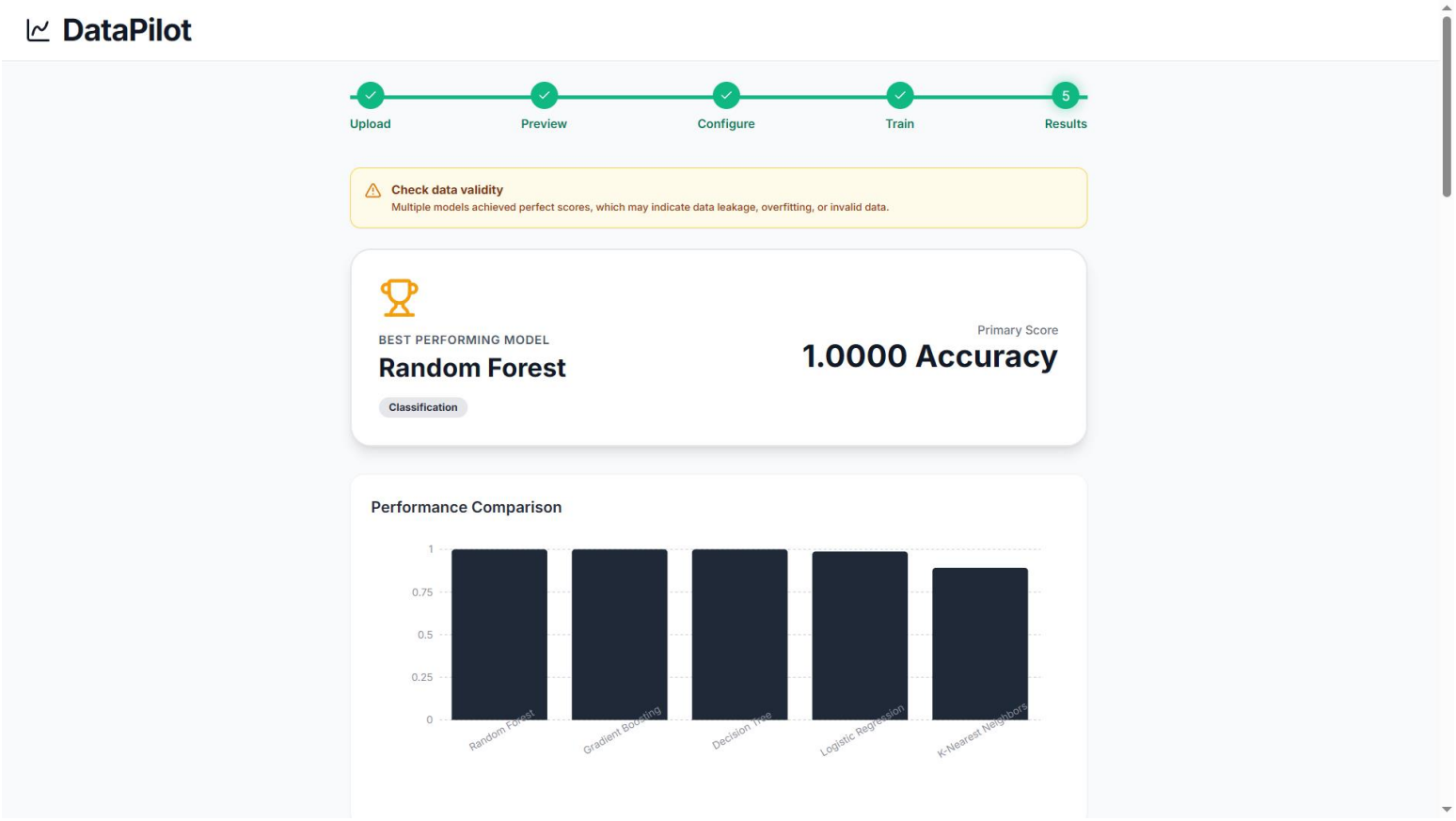


Figure 51 Data Validity Test: Model results

Almost all models achieved perfect scores which confirms that data leakage existed in this environment. In continuation, the confusion matrices appear to follow the perfect score pattern.

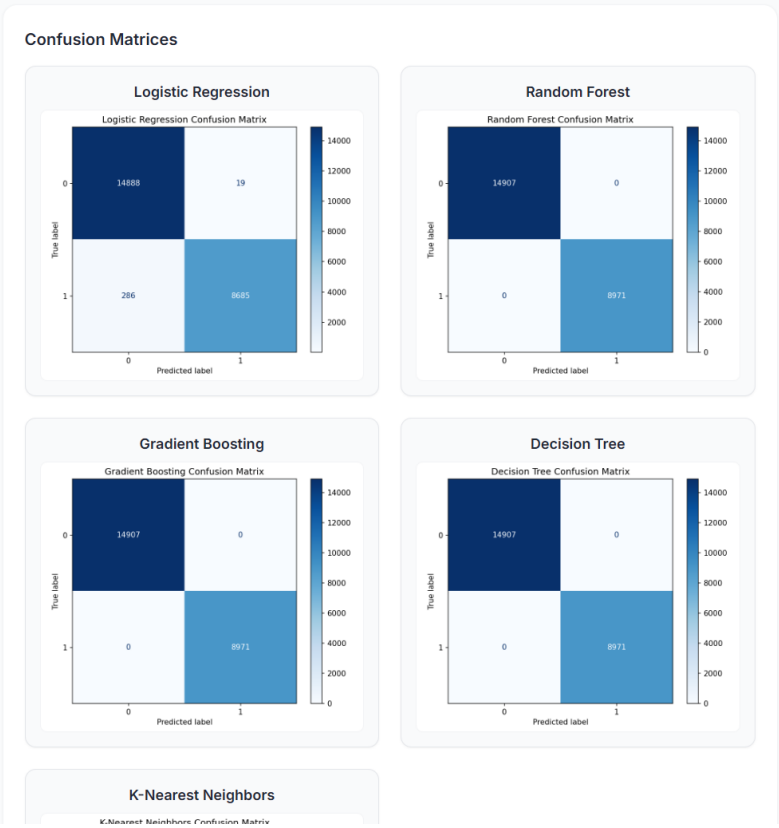


Figure 52 Data Validity Test: Model results - Confusion matrices

3.5.5 Insights of Testing Results

The conducted tests demonstrated the system's analytical capabilities and performance on regression and classification tasks from different datasets. The models included achieved significantly strong results in more than one case. The tree-based models appear to be the strongest based on the conducted tests. The system seems to be able to handle any changes and configurations smoothly as well as identify and present potential issues or behaviors of the uploaded data.

4 Conclusion

Ultimately, the literature research can be concluded with the following findings: The relationship between ML and its branches of AutoML and XAI appears to be significant concerns and aspects of modern analytical environments. Predictive analytics and ML techniques have become essential drivers for pattern extraction, forecasting and supporting data-driven decision-making in multiple domains.

The research that has been covered through this study was initialized as an entry point to the topic of ML and DSS. As the introduced literature became more specific, the increasing importance of automation in ML workflows, through AutoML was covered. The use of such frameworks can reduce complexity in model development and improve understanding of this or any similar process within data science. Meanwhile, the literature review covered challenges that can be introduced with the integration of this automation. The challenges are closely related to transparency, explainability and trust since many systems that use automation because of their

complexity, operate as “black boxes.” Thus, the integration of explainable and human centering approaches is identified as an important requirement for modern DSS.

Based on these theoretical foundations, DataPilot, the proposed system, was implemented as an automated analytical application capable of supporting both regression and classification tasks through a web interface. The system followed a design and workflow approach from research of the industry in order to provide the best possible experience, result explainability and structured workflow for analytical problems. The system is able to organize and automate significant stages of an analytical process including dataset handling, preprocessing, model training, evaluation, visualization and data validity checks.

The tests that took place demonstrated the system's ability to handle the two analytical task types using different datasets and encouraged the creation of conclusions based on model results. In addition, tests regarding special occasions were conducted. These tests included a differentiation of the train/test split of data and the event of data leakage. In all cases, the system was able to identify problematic results and unrealistic model performance while at the same time, functioning correctly in common and unique training cases.

By taking all into consideration, the findings of this study support the idea that automated analytical systems can effectively support DSS environments by improving efficiency, explainability and predictive capabilities for decision support. Nevertheless, emphasis should be put towards the importance of transparency and user understanding within automated processes.

4.1 Objectives Achieved

As a whole, this study aimed to introduce, promote and contribute to the transparency of Human-AI collaboration. This was achieved with research of literature and the presentation of examples, experimentation and conclusions drawn by researchers of this field and its many domains. The integration of ML for decision support and business value is an important aspect that this study covered. Through research of literature, presentation of automated systems and insights granted by testing of implementation, it can be argued that the value of a modern analytical application that supports the automation of an ML workflow, can provide value to a business. With the achievement of these goals, this study has established some starting steps towards additional research that can start from either the review of literature, the covered case studies, or the proposed system.

4.2 Future Work

Future work regarding the covered topic can include the value of Unsupervised learning in automated conditions or the extension of this research to an equivalent one derived from this learning technique. Moreover, further considerations and industry examples of human-AI integration in DSS may increase the reliability and understanding of this topic and its integrations by covering the limits that may be surpassed and those that must not.

Numerous improvements can take place within the proposed system. An important improvement can be the creation of a manual preprocessing interface where users will be able to decide which data gets removed, what missing data should be replaced with, as well as introducing the

capability of feature engineering for stronger analytical solutions. Furthermore, the system's ability to process very large datasets can be enhanced for analyses of greater scale. In terms of explainability, the system could enhance the model results interface with more information regarding model scores on metrics as well as predefined or generated text regarding model results, data behavior and suggestions of future steps that analysts can take. These steps can broaden the research of this field and aid in the creation of a system capable of supporting the analytical process and forthcoming decisions.

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